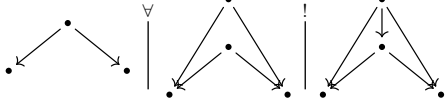


Categories, Allegories

Freyd & Scedrov, all of chapter 1 from §1.4 — definitions, diagrams, theorems

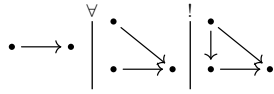
1.4 Cartesian categories

- 1.41 **MONIC PAIR** — $!$ reads “there is at most one extension to”:



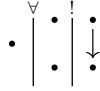
in symbols: $\Box x = \Box y \wedge \forall_{u,v}[(ux = vx) \wedge (uy = vy)] \Rightarrow u = v$.

a single **MONIC** x , written $A \rightarrowtail B$ — variously called monomorphism, mono, injection; here always monic:

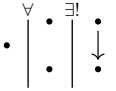


- 1.412 monic family: $\forall_{x \in \mathcal{F}} ux = vx \Rightarrow u = v$. A **TABLE** $\langle T; x_1, \dots, x_n \rangle$ is the **TOP** T with a monic finite sequence out of it; the targets are the **FEET**, the x_i the **COLUMNS**.
- an isomorphism class of tables is a **RELATION**, $\mathcal{Rel}(A_1, \dots, A_n)$ the family of them on given feet. $n = 1$: a **SUBJECT** of A , family $\mathcal{Sub}(A)$. $n = 0$: a **VALUE**, family $\mathcal{Val}$.

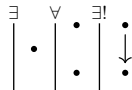
SUB-TERMINATOR — the top of a columnless table:



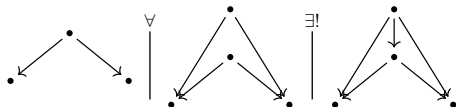
- 1.413 **CONTAINED**, $\langle T; x_i \rangle \subset \langle T'; x'_i \rangle$: some $z : T \rightarrow T'$ with $zx'_i = x_i$ — necessarily unique and monic. Containment pre-orders tables and partially orders relations, mutual containment being isomorphism; $\mathcal{Rel}$, $\mathcal{Sub}$, $\mathcal{Val}$ are posets.
- 1.414 $A \rightarrow B$ is monic in $\mathbf{A}$ iff it is a sub-terminator of $\mathbf{A}/B$. Hence $\mathcal{Sub}_{\mathbf{A}}(B) \simeq \mathcal{Val}_{\mathbf{A}/B}$.
- 1.415 Rel a table on A, B lists, without repetition, the instances of a relation from A to B — the extensional and the categorical notions coincide.
- 1.421 **TERMINATOR** (elsewhere: final, terminal), written 1 , with $p_A : A \rightarrow 1$; a sub-terminator, representing the maximum value, unique up to a unique isomorphism:



a category has a terminator if



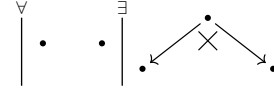
- 1.423 binary **PRODUCT** diagram:



written $A \times B$ with $\ell : A \times B \rightarrow A$, $\varkappa : A \times B \rightarrow B$; $\langle f, g \rangle$ the unique map with $\langle f, g \rangle \ell = f$, $\langle f, g \rangle \varkappa = g$; and $(x \times y) = \langle \ell' x, \varkappa' y \rangle$. $A \times B$ is a table on A, B

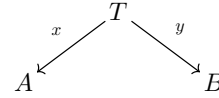
and the maximum of $\mathcal{Rel}(A, B)$, so products are unique up to isomorphism.

a category has binary products if — the cross marking a product diagram:



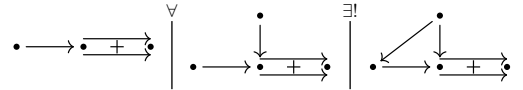
- 1.424 Rel products are the disjoint sums. In a poset, products are greatest lower bounds.
- 1.425 indexed product: P with $\{p_i : P \rightarrow A_i\}_I$ through which every $\{x_i : X \rightarrow A_i\}$ factors uniquely, $zp_i = x_i$; written $\prod_I A_i$. Empty product = terminator; finite products = terminator + binary products. A poset has all finite products iff it is a semi-lattice.

1.426



is a table iff $\langle x, y \rangle : T \rightarrow A \times B$ is monic, so $\mathcal{Rel}(A, B) \simeq \mathcal{Sub}(A \times B)$, generally $\mathcal{Rel}(A_1, \dots, A_n) \simeq \mathcal{Sub}(A_1 \times \dots \times A_n)$, and $\mathcal{Val} = \mathcal{Sub}(1)$.

- 1.428 **EQUALIZER** diagram — the puncture mark removes just the one equation, so the square carrying it need not commute:

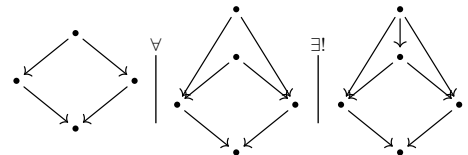


equalizers are monic, so they represent subobjects, and are essentially unique. In $\mathcal{S}$: $\{x \mid fx = gx\}$.

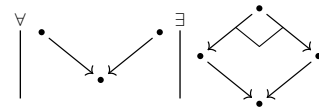
- 1.429 if $e, 1_A$ has an equalizer for every idempotent e , then all idempotents split.

- 1.43 **CARTESIAN CATEGORY**: finite products and equalizers (elsewhere: finitely complete, left-exact).

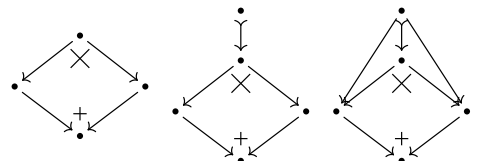
- 1.431 **PULLBACK** diagram — its upper half is a monic pair, and the relation that pair represents is fixed by the lower half:



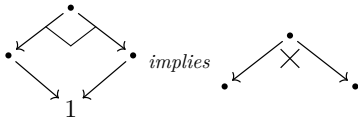
a category has pullbacks if — the corner mark $\lrcorner$ standing for the whole sentence above:



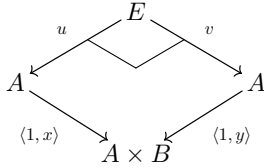
- 1.432 Binary products + equalizers $\Rightarrow$ pullbacks — construct in sequence; the outer diagram on the right is the pullback:



- 1.433 Pullbacks + a terminator $\Rightarrow$ binary products:



- 1.434 *Binary products + pullbacks $\Rightarrow$ equalizers* — given $x, y : A \rightarrow B$, then $u = v$ and u equalizes x, y :

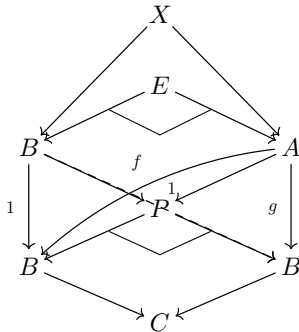


- 1.435–6 combining the last two: *pullbacks and a terminator imply cartesian*. The three lemmas are exhaustive — all sixteen combinations of terminator / binary products / equalizers / pullbacks not forced by them are realized.

- 1.437 a functor preserving finite products and equalizers is a **REPRESENTATION OF CARTESIAN CATEGORIES** such preserve pullbacks. Preserving pullbacks + the terminator suffices; so does preserving finite products + pullbacks of monics.

- 1.438 *A functor reflecting equalizers reflects isomorphisms* ($A \rightarrow B$ is iso iff it equalizes $1_B, 1_B$). From a category with equalizers, an isomorphism-reflecting equalizer-preserving functor is an embedding, hence faithful. A faithful functor preserving terminators / products / equalizers / pullbacks reflects them too.

- 1.439 *With pullbacks, if $A \xrightarrow{f,g} B \rightarrow C$ then f, g have an equalizer* — $P = B \times_C B$ replaces $B \times B$, and $E \rightarrow A$ is the equalizer:



A functor from a cartesian category preserving pullbacks preserves equalizers.

- 1.44 $\Sigma : \mathbf{A}/B \rightarrow \mathbf{A}$ does not preserve terminators — $\mathbf{A}/B$ has the distinguished terminator 1_B , carried to B — and is universal in that: for $T : \mathbf{C} \rightarrow \mathbf{A}$ with $T(1) = B$ there is a unique $T' : \mathbf{C} \rightarrow \mathbf{A}/B$ with $T'(1) = 1_B$ and $T = T'\Sigma$.

specializing to $T = B \times -$ gives the **DIAGONAL FUNCTOR** $\Delta(A) = (B \times A \xrightarrow{\pi} B)$ and the factorization $\mathbf{A} \xrightarrow{\Delta} \mathbf{A}/B \xrightarrow{\Sigma} \mathbf{A}$.

- 1.441 if $\mathbf{A}$ has pullbacks then $\mathbf{A}/B$ is cartesian and Σ preserves pullbacks and equalizers; Σ is always faithful.

- 1.442 the Cayley representation preserves and reflects pullbacks and equalizers; factoring it as $\mathbf{A} \xrightarrow{C'} \mathcal{S}^{|\mathbf{A}|} \xrightarrow{\Sigma} \mathcal{S}$ gives a faithful representation. *Every small cartesian category is faithfully represented in a power of $\mathcal{S}$.*

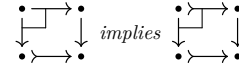
the A -th coordinate of C' is the **REPRESENTABLE FUNCTOR** $(A, -) : B \mapsto (A, B)$, $f \mapsto (A, f)$, i.e. $g \mapsto gf$. *Representable functors from a cartesian category*

are representations of cartesian categories, and are collectively faithful.

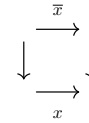
- 1.443 a universal sentence in the pullback / equalizer predicates with a counterexample anywhere has one in $\mathcal{S}$: cut down to a small subcategory, then apply Cayley.

- 1.444 **HORN SENTENCE**: a universally quantified $(P_1 \wedge \dots \wedge P_n) \Rightarrow Q$ with all P_i, Q primitive; for cartesian categories the primitives are the categorical ones plus “is a terminator / product / equalizer”. *Any Horn sentence in the theory of cartesian categories true for $\mathcal{S}$ is true for every cartesian category* — apply a collectively faithful $(A, -)$ to the counterexample.

- 1.45 pullbacks transfer monics:

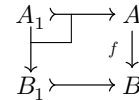


that is, if



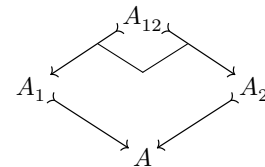
is a pullback and x is monic, then so is $\bar{x}$.

- 1.451 $A_1 \rightharpoonup A$ is an **INVERSE IMAGE** of $B_1 \rightharpoonup B$ along f :



inverse images preserve containment, hence are well defined on subobjects: an order-preserving $f^\# : \mathcal{S}ub(B) \rightarrow \mathcal{S}ub(A)$, making $\mathcal{S}ub(-)$ a contravariant poset-valued functor.

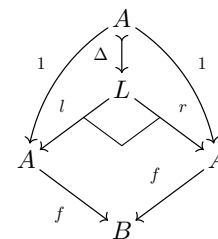
- 1.452 pulling back two monics gives their greatest lower bound in $\mathcal{S}ub(A)$:



so $\mathcal{S}ub(A)$ is a **SEMI-LATTICE** — a poset with finite intersections, the empty one 1_A — and $\mathcal{S}ub(-)$ lands in semi-lattices, inverse images preserving intersections.

- 1.453 **LEMMA** $\mathbf{A}$ cartesian, T preserving pullbacks: T is faithful iff T preserves properness of subobjects — if $A' \rightharpoonup A$ is not iso, neither is $TA' \rightharpoonup TA$.

- 1.454 the kernel-pair of f measures how far f is from monic: the **LEVEL** of f (elsewhere: kernel-pair, congruence). f is monic iff Δ is iso; this monic $\Delta : A \rightarrow L$ is the **DIAGONAL** morphism, representing the diagonal subobject.



- 1.46 the cartesian structure of the prime examples.

- 1.462 for small $\mathbf{A}$, $\mathcal{S}^{\mathbf{A}}$ is cartesian and the **EVALUATION FUNCTORS** $E_A : \mathcal{S}^{\mathbf{A}} \rightarrow \mathcal{S}$, $E_A(F) = F(A)$, are representations of cartesian categories, collectively faithful. Assembled into the forgetful $\mathcal{S}^{\mathbf{A}} \rightarrow \mathcal{S}^{|\mathbf{A}|}$ they

give a faithful representation into a power of $\mathcal{S}$.
Hence $F_1 \rightarrow F_2$ is monic iff every $F_1 A \rightarrow F_2 A$ is.

1.463 H^A denotes $(A, -)$ viewed in $\mathcal{S}^A$. As a right $\mathbf{A}$ -set it is the sub $\mathbf{A}$ -set of $\mathbf{A}$ generated by 1_A : for any right $\mathbf{A}$ -set X and $x \in X$ with $x \square = A$ there is a unique $H^A \rightarrow X$ sending $1_A \mapsto x$, $y \mapsto xy$. So $(H^A, X) \simeq \{x \in X \mid x \square = A\}$, and for the functor F associated to X , $(H^A, F) \simeq F(A)$, naturally — $(H^A, -)$ and E_A are conjugate.

1.464 $(H^A, H^B) \simeq (B, A)$, so each $x : B \rightarrow A$ gives $H^x : H^A \rightarrow H^B$ and a contravariant full embedding $H : \mathbf{A} \rightarrow \mathcal{S}^A$, the **YONEDA REPRESENTATION**.

$\text{Hom} : \mathbf{A}^\circ \times \mathbf{A} \rightarrow \mathcal{S}$ sends A, B to (A, B) ; the composite $\mathbf{A} \times \mathcal{S}^A \xrightarrow{H \times 1} (\mathcal{S}^A)^\circ \times \mathcal{S}^A \xrightarrow{\text{Hom}} \mathcal{S}$ is naturally equivalent to $E : A, F \mapsto F(A)$.

1.465 H_B denotes the contravariant $(-, B)$ viewed in $\mathcal{S}^{A^\circ}$; again $(H_B, F) \simeq F(B)$, giving a covariant full embedding $\mathbf{A} \rightarrow \mathcal{S}^{A^\circ}$, also called the yoneda representation. It preserves and reflects the cartesian predicates.

1.47 $\mathbf{A}$ is *special* if every universal (not merely Horn) sentence in the cartesian predicates true for $\mathcal{S}$ holds in $\mathbf{A}$.

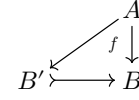
1.471–4 $\mathbf{A}$ *special cartesian category has at most two values*. If every finite sequence of morphisms is preserved-and-reflected by some $T : \mathbf{A} \rightarrow \mathcal{S}$, then $\mathbf{A}$ is special. One-valued $\mathbf{A}$: special iff every $B \times -$ is faithful. Two-valued $\mathbf{A}$, with 0 the proper subobject of 1 : special iff $B \times -$ is faithful for every $B \neq 0$.

1.48 a class $\mathcal{D}$ of monics in a cartesian $\mathbf{A}$ is **DENSE** if it holds the isomorphisms and is closed under composition and pullback. The **RATIONAL CATEGORY** $\mathbf{A}[\mathcal{D}^{-1}]$, with $T_{\mathcal{D}}$, inverts every dense monic and is universal among functors that do.

1.481 $\mathbf{A}[\mathcal{D}^{-1}]$ is cartesian and $T_{\mathcal{D}}$ is a representation of cartesian categories.

1.5 Regular categories

1.51 a subobject $B' \rightarrowtail B$ **ALLOWS** f iff f factors through it:



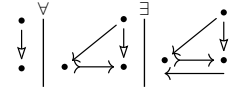
with pullbacks: B' allows f iff $f^\#(B')$ is entire, all of A . A faithful pullback-preserving functor reflects allowance.

the **IMAGE** of f is the smallest subobject allowing f ; a category **HAS IMAGES** if every morphism has one. Then $f : \text{Sub}(A) \rightarrow \text{Sub}(B)$ sends $A' \rightarrowtail A$ to the image of $A' \rightarrowtail A \rightarrow B$.

with pullbacks too, the first adjoint pair of poset functions: $f(A') \subset B'$ iff $A' \subset f^\#(B')$. f is the **LEFT-ADJOINT** of $f^\#$, $f^\#$ the **RIGHT-ADJOINT** of f .

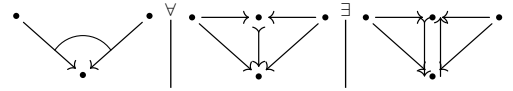
1.511 If $\mathbf{A}$ has images and T is faithful and preserves images, then T reflects images.

1.512 $A \rightarrow B$ is a **COVER**, written $A \twoheadrightarrow B$, if its image is entire — every monic it factors through is invertible:



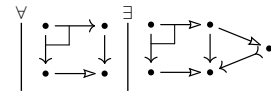
covers are closed under composition and left cancellation ($A \rightarrow B \rightarrow C$ a cover forces $B \rightarrow C$ to be one); a monic cover is an isomorphism.

1.513 $\{A_i \rightarrow B\}$ **COVERS** if no proper subobject of B allows all of them — the arc across the two arrows says the pair covers:



1.514 $\{A_i \rightarrow B\}$ is **EPIC** if its members collectively cancel — a monic family in the opposite category. With equalizers, cover implies epic; the converse is a special property.

1.52 a **REGULAR CATEGORY** is a cartesian category with images in which pullbacks transfer covers — one sentence for both clauses, the right-hand vertex being the image of the right leg:

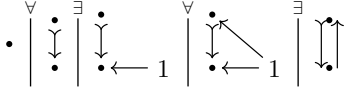


a **PRE-REGULAR CATEGORY** is cartesian, with or without images, and transfers covers by pullback — the strength at which the statements below hold. A **REPRESENTATION OF PRE-REGULAR CATEGORIES** preserves finite products, equalizers and covers, hence whatever images exist.

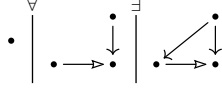
1.521 for small $\mathbf{A}$, $\mathcal{S}^A$ is regular and the evaluation functors are collectively faithful representations of regular categories.

1.522 the **SUPPORT** $\text{Spt}(A)$ is the image of $A \rightarrow 1$; A is **WELL-SUPPORTED** if $\text{Spt}(A) = 1$ — pre-regularly, if $A \rightarrow 1$ covers.

1.523 A is **WELL-POINTED** if all of $1 \rightarrow A$ covers A : no proper subobject of A allows every point.



1.524 P is **PROJECTIVE** if $(P, -)$ preserves covers — every $P \rightarrow B$ lifts through every cover:



Pre-regularly it suffices that every cover targeted at P have a left inverse.

1.525 a pre-regular category is **CAPITAL** if every well-supported object is well-pointed. In a capital pre-regular category the terminator is projective.

1.526 so for capital pre-regular $\mathbf{A}$ the functor Γ represented by 1 is a representation of pre-regular categories.

1.53 **THE SLICE LEMMA** factor $(B \times -)$ as $\mathbf{A} \xrightarrow{\Delta} \mathbf{A}/B \xrightarrow{\Sigma} \mathbf{A}$, Δ the diagonal functor [1.44], $\Delta(1) = (B \xrightarrow{1} B)$. If $\mathbf{A}$ is (pre-)regular so is $\mathbf{A}/B$; Δ is a representation of pre-regular categories; Δ is faithful iff B is well-supported.

1.531 $\Sigma : \mathbf{A}/B \rightarrow \mathbf{A}$ preserves and reflects covers and pullbacks and carries factorizations back, so images in $\mathbf{A}$ give images in $\mathbf{A}/B$.

1.535 *Diagonal* also names the morphism $A \rightarrow \Pi_I A$ with identity coordinates; Δ is not that functor but is conjugate to it.

1.54 **THE CAPITALIZATION LEMMA** let $\mathcal{C}$ have small pre-regular categories as objects and faithful representations as morphisms.

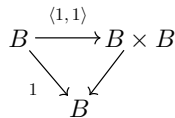
1.541 $\mathcal{C}$ meets the *equivalence condition* if $\mathbf{A} \in \mathcal{C}$ and an equivalence $\mathbf{A} \rightarrow \mathbf{B}$ force $\mathbf{B}, (\mathbf{A} \rightarrow \mathbf{B}) \in \mathcal{C}$; the *slice condition* if $\mathbf{A} \in \mathcal{C}$ and B well-supported force $\mathbf{A}/B, \Delta \in \mathcal{C}$; the *union condition* if a **DIRECTED UNION** $\mathbf{A} = \bigcup \mathcal{F}$ of $\mathcal{C}$ -subcategories with $\mathcal{C}$ -inclusions has $\mathbf{A}, (\mathbf{A}' \subset \mathbf{A}) \in \mathcal{C}$.

1.542 all three hold for the small pre-regular and the small regular categories, the slice condition being [1.53].

1.543 Under the three conditions every $\mathbf{A} \in \mathcal{C}$ admits a capital $\underline{\mathbf{A}} \in \mathcal{C}$ with $\mathbf{A} \rightarrow \underline{\mathbf{A}}$ faithful in $\mathcal{C}$.

1.544 to read $\mathbf{A}$ inside $\mathbf{A}/B$, pass to the **INFLATION** $\mathbf{A}'$ — objects finite sequences of objects, binary product concatenation — giving strict cancellation $B \times A = B \times A' \implies A = A'$, then replace the image of Δ by $\mathbf{A}$.

1.545 $\mathbf{A} \subset \mathbf{A}^*$ is a **RELATIVE CAPITALIZATION** if every proper $B' \rhd B$ of $\mathbf{A}$, B well-supported, fails to allow some $x : 1 \rightarrow B$ of $\mathbf{A}^*$. Iterated union gives $\underline{\mathbf{A}}$; each slicing step supplies ΔB with its generic point, which $\Delta B'$ cannot allow:



1.547 choice-free variant: the rational category [1.48] of pairs $\langle A, F \rangle$, F a finite set of morphisms to distinct well-supported targets — a directed union of slices $\mathbf{A}/\Pi U$.

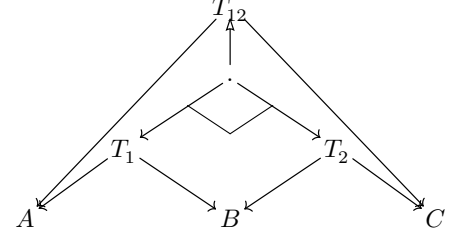
1.55 **THE HENKIN-LUBKIN THEOREM** Every small pre-regular category is faithfully represented in a power of the category of sets: $T_B = \mathbf{A} \xrightarrow{\Delta} \mathbf{A}/B \xrightarrow{\Gamma} \mathbf{S}$; for regular $\mathbf{A}$ the $\{T_U\}_{U \subset 1}$ already suffice.

1.551 Every Horn sentence in the defining predicates of regular categories true for the category of sets is true for every regular category.

1.552 a pre-regular category is **SPECIAL** if every universally quantified sentence in the relevant predicates true for $\mathcal{S}$ holds in it; the conditions of [1.47] stay necessary. *Special* iff $A \rightarrow U \rhd 1$ forces $A \rightarrow U$ or $U \rhd 1$ to be an isomorphism

equivalently: one-valued ($|\pi_0 \mathbf{A}| = 1$), or two-valued ($|\pi_0 \mathbf{A}| = 2$) with every object well-supported or isomorphic to 0 , the unique proper sub-terminator.

1.56 tables T_1 on A, B and T_2 on B, C : a table T_{12} on A, C is a **COMPOSITION** if there exists



cartesian + images builds one exactly so; shrinking either factor shrinks the composite, giving an order-preserving $\mathcal{R}el(A, B) \times \mathcal{R}el(B, C) \rightarrow \mathcal{R}el(A, C)$. *Caution*: not necessarily associative [1.569].

1.561 the **RECIPROCAL** R° of R from A to B twists the columns: $R^\circ = R$, $(RS)^\circ = S^\circ R^\circ$. Reciprocation reverses composition, preserves the ordering.

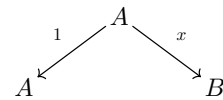
1.562 $\mathcal{R}el(A, B) \simeq \mathcal{S}ub(A \times B)$ makes $\mathcal{R}el(A, B)$ a semi-lattice: $(R \cap S)T \subset RT \cap ST$, $(R \cap S)^\circ = S^\circ \cap R^\circ$.

1.563 F preserving the cartesian structure and images induces $\mathcal{R}el(A, B) \rightarrow \mathcal{R}el(FA, FB)$ preserving composition, reciprocation and intersection; faithful F reflects them.

Hence a Horn sentence on maps and relations — the regular predicates, plus composition, reciprocation and intersection — true of binary relations between sets is true in every regular category.

in particular $R(ST) = (RS)T$, and the **MODULAR IDENTITY** $RS \cap T \subset (R \cap TS)S$. Every containment is an equation: $R \subset S$ iff $R = R \cap S$.

1.564 $\mathcal{R}el(\mathbf{A})$ is the category of relations; the **GRAPH** of x is tabulated by

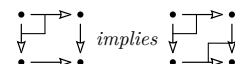


giving $\mathbf{A} \rightarrow \mathcal{R}el(\mathbf{A})$. Treat $\mathbf{A}$ as a subcategory: **MAP** means a morphism of $\mathcal{R}el(\mathbf{A})$ lying in $\mathbf{A}$, and $\langle T; x, y \rangle$ tabulates one iff x is an isomorphism.

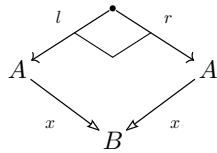
R is **ENTIRE** iff $1 \subset RR^\circ$, **SIMPLE** iff $R^\circ R \subset 1$. For R tabulated by x, y : x covers iff R is entire, x is monic iff R is simple. So R is a map iff entire and simple — hence as soon as every representation $F : \mathbf{A} \rightarrow \mathcal{S}$ makes $F(R)$ one.

1.565 a **PUSHOUT** is a pullback in the opposite category; the corner mark moves to the far vertex of the square, C .

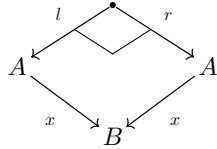
In a regular category a pullback of covers is a pushout — all four edges then cover, the mediating map being the relation $(x^\circ u) \cap (y^\circ v)$, a map because it is one in $\mathcal{S}$:



1.566 a **COEQUALIZER** — an equalizer in the opposite category — is always a cover. In a regular category every cover is a coequalizer — of its own level:

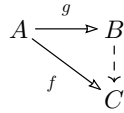


- 1.567 an endo-relation E is an **EQUIVALENCE RELATION** if $1 \subset E$, $E^\circ \subset E$, $EE \subset E$. Levels are equivalence relations:



E is **EFFECTIVE** if it is the level of some map; an **EFFECTIVE REGULAR CATEGORY** is one where every equivalence relation is effective. The metatheorem [1.551] extends to these, $\mathcal{S}$ being effective.

- 1.568 covers out of A are preordered by $f \leq g$ iff f factors through g :



the associated poset is $Quot(A)$, of **QUOTIENT-OBJECTS** — not the dual of $Sub(A)$. By [1.566] a faithful order-reversing functor sends it to the poset of equivalence relations on A .

- 1.569 For $\mathbf{A}$ cartesian with images, composition of relations is associative iff $\mathbf{A}$ is regular — x covers iff $x^\circ x = 1$; if z fails to cover, $(yx^\circ)x$ is not even a map.
- 1.56(10) x is a **CONSTANT MORPHISM** if $yx = y'x$ for all $y, y' : C \rightarrow A$. In a regular category its image is a sub-terminator.
- 1.56(11) In a regular category an object is projective iff every entire relation out of it contains a map.
- 1.57 an object is **CHOICE** if every entire relation targeted at it contains a map. Every object is projective iff every object is choice; an **AC REGULAR CATEGORY** is a regular category with either — equivalently, a cartesian category where every morphism is a left-invertible followed by a monic.
- 1.571 if $\mathbf{A}$ is cartesian and every $x : A \rightarrow B$ admits $e : A \rightarrow A$ with $e^2 = e$, $ex = x$, then e and x share a level and $\mathbf{A}$ is AC regular — the image of x appears as a subobject of A .
- 1.572 the category $\mathbf{R}$ of recursive functions on $0, 1, 2, \dots, \omega$ is AC regular but not effective: a left inverse would choose representatives, forcing a recursively enumerable equivalence relation to be recursive.

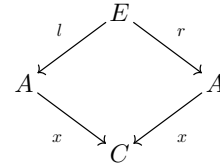
- 1.58 a **BICARTESIAN CATEGORY** is both cartesian and **COCARTESIAN**, the latter meaning the opposite category is cartesian. Dual to a terminator is a **COTERMINATOR**, written 0 ; dual to a product a **COPRODUCT**, written $A + B$.

in a cartesian category an object 0 whose incoming morphisms are all isomorphisms is a coterminator, then called a **STRICT COTERMINATOR** — agreeing with the earlier 0 [1.474, 1.552].

- 1.581 between regular cocartesian categories a representation of bicartesian categories is a representation of regular categories: it preserves coequalizers, hence covers [1.566].

- 1.582 for bicartesian $\mathbf{A}$ regularity is a pair of Horn sentences in the bicartesian predicates — the image of x is $A \twoheadrightarrow C \twoheadrightarrow B$, with $A \twoheadrightarrow C$ the coequalizer of the level of x .

- 1.583 effectiveness likewise: for l, r tabulating an equivalence relation and x their coequalizer, E is effective iff



is a pullback.

- 1.584 $\mathbf{A}/B$ is cocartesian and Σ a faithful representation of cocartesian categories.
- 1.586 for small $\mathbf{A}$, $\mathcal{S}^{\mathbf{A}}$ is cocartesian with collectively faithful evaluation functors, hence effective regular.
- 1.587 no elementary characterization of the bicartesian categories obeying every Horn sentence true for $\mathcal{S}$: $1 + \mathbb{N} \xrightarrow{(s)} \mathbb{N}$ iso plus $\mathbb{N} \rightarrow 1$ a coequalizer of $1, s$ pins down $\mathbb{N}, 0, s$, then addition and multiplication; the equalizer of two polynomials is a coterminator iff the equation is unsolvable, so those Horn sentences encode the diophantine problems and are not recursively enumerable.

1.6 Pre-logoi

- 1.6 A **PRE-LOGOS**: a regular category in which $\mathcal{S}ub(A)$ is a lattice, not just a semi-lattice, for each A , and $f^\# : \mathcal{S}ub(B) \rightarrow \mathcal{S}ub(A)$ is a lattice homomorphism for each $f : A \rightarrow B$.

Elementary form: each pair of subobjects has a union, preserved under inverse image.

Equivalently: a cartesian category with images in which pullbacks transfer finite covers. For a cover $\{B_i \rightarrow B\}_{i=1}^n$ and a map $A \rightarrow B$ there exist

$$\begin{array}{ccc} A_i & \longrightarrow & A \\ \downarrow & & \downarrow \\ B_i & \longrightarrow & B \end{array}$$

$i = 1, \dots, n$, with $\{A_i \rightarrow A\}_{i=1}^n$ a cover. Their existence forces the pullback squares to be such.

- 1.61 0 = the minimal subobject of 1 ; $p_A^\#(0)$ = the minimal subobject of A . Any morphism to 0 is an isomorphism. Each A receives exactly one $0 \rightarrow A$: 0 is a coterminator.

A pre-logos is degenerate iff it is one-valued, $0 \simeq 1$.

- 1.611 Hence a pre-logos is a cartesian category with images and three sentences. First, 0 exists and every morphism to it is invertible:

$$\begin{array}{c} \exists \\ \downarrow \\ \cdot \\ \downarrow \\ \exists \end{array} \quad \begin{array}{c} \forall \\ \downarrow \\ \cdot \\ \downarrow \\ \exists \end{array} \quad \begin{array}{c} \exists \\ \downarrow \\ \cdot \\ \downarrow \\ \exists \end{array} \quad \begin{array}{c} \exists \\ \downarrow \\ \cdot \\ \downarrow \\ \exists \end{array}$$

second, any two morphisms with a common target have a union — the arc says the pair covers it:

$$\begin{array}{c} \forall \\ \downarrow \\ \cdot \\ \downarrow \\ \exists \end{array} \quad \begin{array}{c} \forall \\ \downarrow \\ \cdot \\ \downarrow \\ \exists \end{array} \quad \begin{array}{c} \forall \\ \downarrow \\ \cdot \\ \downarrow \\ \exists \end{array}$$

third, inverse image preserves unions — the squares of 1.6 again, now with both pairs covering:

$$\begin{array}{c} \forall \\ \downarrow \\ \cdot \\ \downarrow \\ \exists \end{array} \quad \begin{array}{c} \forall \\ \downarrow \\ \cdot \\ \downarrow \\ \exists \end{array} \quad \begin{array}{c} \forall \\ \downarrow \\ \cdot \\ \downarrow \\ \exists \end{array}$$

- 1.612 For monic $f : A \rightarrow B$ the inverse image sends $B' \subset B$ to $A \cap B'$.

Every such $f^\#$ preserves binary unions iff $\mathcal{S}ub(B)$ is a **DISTRIBUTIVE LATTICE**: $A \cap (B_1 \cup B_2) = (A \cap B_1) \cup (A \cap B_2)$.

- 1.613 A poset viewed as a category is cartesian iff it is a semi-lattice (intersections are products, equalizers hold by default); a semi-lattice is regular, the only covers being identities. A semi-lattice is a pre-logos iff it is a distributive lattice.

- 1.614 A **REPRESENTATION OF PRE-LOGOI**: a functor between pre-logoi preserving the cartesian structure, images, and finite unions, the empty union included.

- 1.615 In a bicartesian category with images the union of $x_1 : A_1 \rightarrow A$, $x_2 : A_2 \rightarrow A$ is the image of $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} : A_1 + A_2 \rightarrow A$.

Building on [1.582], a finite set of Horn sentences in the bicartesian predicates holds for a bicartesian category iff it is a pre-logos.

- 1.616 $\mathcal{R}el(A, B) \simeq \mathcal{S}ub(A \times B)$ is a distributive lattice, so relations carry $R \cup S$.

$\mathcal{S}ub(A \times B)$ distributive	$R \cap (S \cup T) = (R \cap S) \cup (R \cap T)$
reciprocation	$(R \cup S)^\circ = S^\circ \cup R^\circ$
$yS = (y \times 1)^\#(S)$, inverse images	$y(S \cup T) = yS \cup yT$
$x^\circ(yS) = (x \times 1)(yS)$, direct images	$R(S \cup T) = RS \cup RT$
reciprocate the line above	$(S \cup T)R' = SR' \cup TR'$

- 1.62 **PASTING LEMMA** For subobjects A_1, A_2 of A in a pre-logos,

$$\begin{array}{ccc} A_1 \cap A_2 & \longrightarrow & A_2 \\ \downarrow & & \downarrow \\ A_1 & \longrightarrow & A_1 \cup A_2 \end{array}$$

is a pushout — the mediating map out of $A_1 \cup A_2$ is unique because A_1, A_2 cover it.

Restated: a pullback whose two morphisms into the lower right corner cover is a pushout.

$$\begin{array}{ccc} \begin{array}{c} \cdot \\ \downarrow \\ \cdot \end{array} & \xrightarrow{\quad} & \begin{array}{c} \cdot \\ \downarrow \\ \cdot \end{array} \\ \downarrow & \searrow & \downarrow \\ \begin{array}{c} \cdot \\ \downarrow \\ \cdot \end{array} & \xrightarrow{\quad} & \begin{array}{c} \cdot \\ \downarrow \\ \cdot \end{array} \end{array} \text{ implies } \begin{array}{ccc} \begin{array}{c} \cdot \\ \downarrow \\ \cdot \end{array} & \xrightarrow{\quad} & \begin{array}{c} \cdot \\ \downarrow \\ \cdot \end{array} \\ \downarrow & \searrow & \downarrow \\ \begin{array}{c} \cdot \\ \downarrow \\ \cdot \end{array} & \xrightarrow{\quad} & \begin{array}{c} \cdot \\ \downarrow \\ \cdot \end{array} \end{array}$$

- 1.621 If $A_1 \cap A_2 = 0$ and $A_1 \cup A_2 = A$ then A is a coproduct of A_1, A_2 — the special case of the square above.

- 1.623 A **POSITIVE PRE-LOGOS**: a pre-logos in which every pair A, B admits

$$\begin{array}{ccc} 0 & \longrightarrow & B \\ \downarrow & \searrow & \downarrow \\ A & \longrightarrow & C \end{array}$$

Choosing C with $A \cup B = C$, a positive pre-logos has coproducts.

- 1.624 In a positive pre-logos any $f : A \rightarrow B_1 + B_2$ yields a decomposition, $A_i = f^\#(B_i)$:

$$\begin{array}{ccc} A & \xrightarrow{\quad} & A_1 + A_2 \\ \downarrow f & \searrow & \downarrow f_1 + f_2 \\ & & B_1 + B_2 \end{array}$$

- 1.625 For $\mathbf{A}, \mathbf{B}$ positive pre-logoi and $T : \mathbf{A} \rightarrow \mathbf{B}$ a representation of regular categories: T is a representation of pre-logoi iff it preserves disjoint unions.

- 1.63 $\Sigma : \mathbf{A}/\mathbf{B} \rightarrow \mathbf{A}$ yields $\mathcal{S}ub_{\mathbf{A}/\mathbf{B}}(-) \simeq \mathcal{S}ub_{\mathbf{A}}(\Sigma(-))$, respecting inverse images when $\mathbf{A}$ has pullbacks, and unions in $\mathbf{A}$ give unions in $\mathbf{A}/\mathbf{B}$.

If $\mathbf{A}$ is a (positive) pre-logos so is $\mathbf{A}/\mathbf{B}$, and $\Delta : \mathbf{A} \rightarrow \mathbf{A}/\mathbf{B}$ is a representation of pre-logoi.

The capitalization lemma [1.543] applies: any (positive) pre-logos is faithfully representable in a capital (positive) pre-logos.

- 1.631 $A_1 \subset A$ is a **COMPLEMENTED SUBOBJECT** if some $A_2 \subset A$ has $A_1 \cap A_2 = 0$, $A_1 \cup A_2 = A$ [1.621]; distributivity of $\mathcal{S}ub(A)$ makes A_2 unique.

In a positive pre-logos a complemented subobject of a projective object is projective. With [1.525]: the **COMPLEMENTED SUBTERMINATORS** — complemented subobjects of 1 — in a capital pre-logos are projective.

- 1.632 $\mathcal{G} \subset |\mathbf{A}|$ is a **GENERATING SET** if $\{(G, -)\}_{G \in \mathcal{G}}$ collectively yields an embedding; $\mathcal{B} \subset |\mathbf{A}|$ is a **BASIS** if $\{(B, -)\}_{B \in \mathcal{B}}$ is collectively faithful. Both elementary.

For cartesian $\mathbf{A}$, $\mathcal{B}$ is a basis iff every proper $A' \rightarrow A$ admits

$$\begin{array}{ccc} B' & \longrightarrow & B \\ \downarrow & \searrow & \downarrow \\ A' & \longrightarrow & A \end{array}$$

with $B \in \mathcal{B}$ and $B' \rightarrow B$ proper [1.442].

- 1.633 A positive pre-logos is capital iff the complemented subterminators are projective and form a basis.
- 1.634 $\mathcal{F} \subset P$ is a **PRE-FILTER** if non-empty and any $x, y \in \mathcal{F}$ have $z \in \mathcal{F}$ with $z \leq x, z \leq y$; a **FILTER** if also an updeal.

For $\mathcal{F}$ a pre-filter in $\mathcal{V}\mathcal{a}\mathcal{L}_{\mathbf{A}}$: $T_{\mathcal{F}}(A)$ has elements named by $x : U \rightarrow A, U \in \mathcal{F}$, with x and $y : V \rightarrow A$ naming the same element when

$$\begin{array}{ccc} W & \longrightarrow & U \\ \downarrow & & \downarrow x \\ V & \xrightarrow{y} & A \end{array}$$

for some $W \in \mathcal{F}, W \subset U, W \subset V$.

$T_{\mathcal{F}}$ preserves finite products and equalizers; it preserves covers when the elements of $\mathcal{F}$ are projective.

In a pre-logos $T_{\mathcal{F}}$ preserves disjoint unions iff $0 \notin \mathcal{F}$, and $U_1 + U_2 \in \mathcal{F}$ implies $U_1 \in \mathcal{F}$ or $U_2 \in \mathcal{F}$.

- 1.635 **THE REPRESENTATION THEOREM FOR PRE-LOGOI** Every small positive pre-logos is faithfully representable in a power of the category of sets. ‘Positive’ is removable, every pre-logos being faithfully representable in a positive one [2.217].

In a capital positive pre-logos a representative set $\mathcal{B}$ of complemented subterminators, as a sublattice of $\mathcal{S}\mathcal{u}\mathcal{b}$, is distributive with every element complemented — a **BOOLEAN ALGEBRA**.

Every pre-filter $\mathcal{F} \subset \mathcal{B}$ extends, by the axiom of choice, to an **ULTRA-FILTER** $\bar{\mathcal{F}}$: a maximal pre-filter with $0 \notin \bar{\mathcal{F}}$, necessarily a filter.

For $\bar{\mathcal{F}}$ an ultra-filter, $T_{\bar{\mathcal{F}}}$ is a representation of pre-logoi. Collectively faithful, because $\mathcal{B}$ is a basis.

- 1.636 Any Horn sentence in the predicates of pre-logoi true for the category of sets is true for all positive pre-logoi — ‘positive’ again removable [2.217].
- 1.637 A special pre-logos satisfies every universal sentence in the predicates of pre-logoi satisfied by $\mathcal{S}$. A pre-logos is a special regular category iff two-valued.
- A pre-logos is special iff $(A' \times B) \cup (A \times B')$ is a proper subobject of $A \times B$ for every pair of proper $A' \subset A, B' \subset B$.

- 1.64 A **BOOLEAN PRE-LOGOS**: a pre-logos in which $\mathcal{S}\mathcal{u}\mathcal{b}(A)$ is a boolean algebra for all A . $\mathcal{S}\mathcal{u}\mathcal{b}(A)$ being distributive, only complements need be required, and they are then unique:

$$\begin{array}{c} \forall \\ \left| \begin{array}{c} \bullet \\ \downarrow \\ \bullet \end{array} \right| \begin{array}{c} \exists \\ \left| \begin{array}{c} \bullet \\ \downarrow \\ \bullet \end{array} \right| \end{array} \end{array} \quad \begin{array}{c} 0 \\ \left| \begin{array}{c} \bullet \\ \downarrow \\ \bullet \end{array} \right| \end{array} \quad \begin{array}{c} \bullet \\ \downarrow \\ \bullet \end{array}$$

- 1.641 A representation of pre-logoi between boolean pre-logoi preserves the boolean structure automatically: each $\mathcal{S}\mathcal{u}\mathcal{b}(A) \rightarrow \mathcal{S}\mathcal{u}\mathcal{b}(TA)$ preserves complements and is a map of boolean algebras.
- 1.644 Any power of $\mathcal{S}$ is a boolean capital positive pre-logos; confuse $\mathcal{V}\mathcal{a}\mathcal{L}_{\mathcal{S}I}$ with the power set of I .
- For an ultra-filter $\mathcal{F} \subset \mathcal{V}\mathcal{a}\mathcal{L}_{\mathcal{S}I}$, $T_{\mathcal{F}}$ is an **ULTRA-PRODUCT FUNCTOR** and the composite an **ULTRA-POWER FUNCTOR**. Both are representations of boolean pre-logoi: $\mathcal{S} \xrightarrow{\Delta} \mathcal{S}^I \xrightarrow{T_{\mathcal{F}}} \mathcal{S}$.
- 1.645 For $T : \mathbf{A} \rightarrow \mathbf{B}$ a representation of boolean pre-logoi: $\mathcal{K}er(T) = \{U \subset 1 \mid T(U) = 0\}$, the values killed by T ; $\mathcal{P}rop(A' \subset A)$, the *properness*, is the support of the complement of A' .
- $A' \subset A$ is proper iff $\mathcal{P}rop(A' \subset A) \neq 0$; $TA' \subset TA$ is proper iff $\mathcal{P}rop(A' \subset A) \notin \mathcal{K}er(T)$; T is faithful iff $\mathcal{K}er(T) = 0$.

$1 \in \mathcal{K}er(T)$ iff $\mathbf{B}$ is degenerate, so $\mathbf{A}$ two-valued and $\mathbf{B}$ non-degenerate forces T faithful — every ultra-power functor is faithful. $\mathcal{K}er(T_{\mathcal{F}})$ is the complement of $\mathcal{F}$ in $\mathcal{V}\mathcal{a}\mathcal{L}_{\mathbf{A}}$.

- 1.646 Two propositions with the same proof: every small special cartesian category, and every small special positive pre-logos, is faithfully representable in the category of sets.
- 1.647 For proper $A_i \subset B_i$ with complement A'_i ($i = 1, 2$), the complement of $(A_1 \times B_2) \cup (B_1 \times A_2)$ in $B_1 \times B_2$ is $A'_1 \times A'_2$. Hence, by [1.637], a boolean pre-logos is special iff it is two-valued.
- 1.648 If an ultra-power functor preserves coequalizers — one built from $\mathbb{N}, 0, s$ [1.587] suffices — then $\mathcal{F}$ meets every countable partitioning of I : such an ultra-filter is a **COMPLETE MEASURE**.

The **ATOMIC MEASURES** are those of the form $\{j \mid i \leq j\}$; for $\mathcal{F}$ atomic, $T_{\mathcal{F}}$ is conjugate to a projection and the ultra-power functor to the identity functor.

- 1.65 A **PRE-TOPOS**: an effective positive pre-logos. Every pre-logos is faithfully represented in a pre-topos [2.217].

- 1.651 **THE AMALGAMATION LEMMA** Monics with a common source $x : A \rightarrowtail B, y : A \rightarrowtail C$ in a pre-topos extend to

$$\begin{array}{ccc} A & \xrightarrow{y} & C \\ x \downarrow & & \downarrow \bar{y} \\ B & \xrightarrow{\bar{y}} & D \end{array}$$

— take D with level $E = l^\circ x^\circ y r \cup 1 \cup r^\circ y^\circ x l$ on $B + C$, for l, r the coproduct injections. The square is a pullback, hence by [1.62] a pushout.

- 1.652 In a pre-topos covers coincide with epics and monics coincide with cocovers — every monic is an equalizer, so the image of an epic is entire.
- 1.653 For $A \rightarrow B$ and monic $y : A \rightarrowtail C$ in a pre-topos there exist

$$\begin{array}{ccc} A & \xrightarrow{y} & C \\ \downarrow & & \downarrow \\ B & \xrightarrow{\quad} & D \end{array}$$

— factor $A \rightarrow B$ through its image and stack the square with [1.651].

- 1.654 A pre-topos need not be cocartesian. If it is, the last two sections show its opposite category is regular.
- 1.655 If $\mathbf{A}, \mathbf{B}$ are pre-topoi and $T : \mathbf{A} \rightarrow \mathbf{B}$ preserves 0, pushouts, finite products and monics, then T is a bicartesian representation.

- 1.657 In an effective regular category, for R tabulated by

$$\begin{array}{ccc} & T & \\ x \swarrow & & \searrow y \\ A & & A \end{array}$$

and $f : A \rightarrow B$ a coequalizer of x, y : ff° is the minimal equivalence relation containing R .

A pre-topos is cocartesian iff for every endo-relation R there is a minimal equivalence relation containing R . A functor between bicartesian pre-topoi preserves the bicartesian structure iff it is a representation of pre-logoi preserving minimal equivalence relations.

- 1.658 An object $\langle 1, 1 \rangle : A \rightarrowtail A \times A$ has a complement. Every

object of a pre-topos is decidable iff the pre-topos is boolean — pullbacks of complemented subobjects are complemented.

1.66 Choice objects [1.57]: in any regular category a subobject of a choice object is choice, and a quotient of a choice object is choice — a quotient $x : A \rightarrow B$ [1.568] is also a subobject of A , via a map contained in x° .

1.661 In a regular category finite products of choice objects are choice — an entire relation targeted at a terminator is already a map.

1.662 In a pre-topos: (1) binary coproducts of choice objects are choice; (2) $1 + 1$ is choice; (3) the pre-topos is boolean — are equivalent.

(2) restated by $\mathcal{R}el(B, 1 + 1) \simeq \mathcal{S}ub(B) \times \mathcal{S}ub(B)$: an entire relation $B \rightarrow 1 + 1$ is a pair of subobjects covering B , a map $B \rightarrow 1 + 1$ a partition of B — so $X \cup Y = B$ in $\mathcal{S}ub(B)$ gives $X' \subset X, Y' \subset Y$ with $X' \cup Y' = B, X' \cap Y' = 0$.

1.7 Logoi

1.7 a **LOGOS** is a regular category in which every $\mathcal{S}ub(A)$ is a lattice and every $f^\# : \mathcal{S}ub(B) \rightarrow \mathcal{S}ub(A)$ has a right adjoint $f^{\#\#} : \mathcal{S}ub(A) \rightarrow \mathcal{S}ub(B)$: $f^\#(B') \subset A'$ iff $B' \subset f^{\#\#}(A')$.

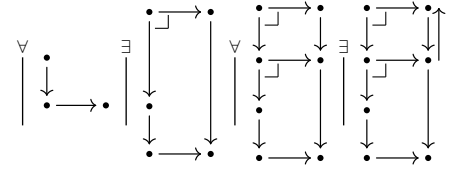
restated: $f^{\#\#}(A')$ is the maximal subobject $B' \subset B$ with $f^\#(B') \subset A'$.

1.71 in $\mathcal{S}$, $f^{\#\#}(A') = \{b \in B \mid \forall_a[f(a) = b \Rightarrow a \in A']\}$. In a boolean pre-logos $f^{\#\#}(A') = \neg(f(\neg A'))$, since $f^\#$ preserves complements.

1.711 a logos is a pre-logos: $f^\#$ preserves every union that exists, since $f^\#(\cup B_i) \subset A'$ iff every $f^\#(B_i) \subset A'$.

1.712 **LOCALLY COMPLETE**: every $\mathcal{S}ub(A)$ is a complete lattice. A locally complete regular category whose $f^\#$ preserve all unions is a logos, with $f^{\#\#}(A') = \cup \{B_i \mid f^\#(B_i) \subset A'\}$.

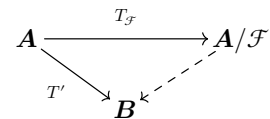
1.714 the axiom drawn — the corner marks make the squares pullbacks, so each left vertex is the inverse image of the right one; the last panel returns $B'' \subset B'$, so no B'' properly containing B' has $f^\#(B'') \subset A'$.



1.73 for a representation of logoi $T : \mathbf{A} \rightarrow \mathbf{B}$, $\mathcal{F}il(T) \subset \mathcal{V}al_{\mathbf{A}}$ collects the values U with $T(U) = 1$; it is a filter.

with $p_A : A \rightarrow 1$: $A' \subseteq A$ iff $p_A^{\#\#}(A') \subseteq 1$, and $T(A') \subseteq T(A)$ iff $p_A^{\#\#}(A') \notin \mathcal{F}il(T)$. T is faithful iff $\mathcal{F}il(T)$ is trivial, $\mathcal{F}il(T) = \{1\}$.

1.731 every filter $\mathcal{F} \subset \mathcal{V}al_{\mathbf{A}}$ gives a logos $\mathbf{A}/\mathcal{F}$ and a representation $T_{\mathcal{F}}$ with $\mathcal{F}il(T_{\mathcal{F}}) = \mathcal{F}$, universal among T' with $\mathcal{F} \subset \mathcal{F}il(T')$:



the mediator is unique up to conjugation, and faithful iff $\mathcal{F} = \mathcal{F}il(T')$.

cartesian / regular / pre-logos $\mathbf{A}$ give cartesian / regular / pre-logos $\mathbf{A}/\mathcal{F}$.

1.732 logoi meet the capitalization lemma's union, equivalence and slice conditions: Σ matches $\mathcal{S}ub_{\mathbf{A}/A}$ with $\mathcal{S}ub_{\mathbf{A}}(\Sigma(_))$ and preserves inverse images, so double-sharps descend to $\mathbf{A}/A$; Δ preserves them because $A \times (_)$ does.

1.733 A in a pre-logos is **COPRIME** if $(A, _)$ preserves finite unions: any finite collection of subobjects of A whose union is A contains A . A is **CONNECTED** if it has exactly two complemented subobjects.

a **FOCAL LOGOS** is a logos whose terminator is a coprime projective, i.e. one for which $\Gamma = (1, _)$ is a representation of pre-logoi. a positive pre-logos is focal iff its terminator is a connected projective.

1.734 $\mathbf{A} \rightarrow \mathbf{F}$ is a **FOCAL REPRESENTATION** if $\mathbf{F}$ is focal, i.e. $\mathbf{A} \rightarrow \mathbf{F} \xrightarrow{\Gamma} \mathcal{S}$ is a representation of pre-logoi.

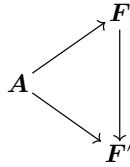
any small (positive) logos has a small, collectively faithful family of focal representations.

a logos may be faithfully represented in a single focal logos iff its terminator is coprime. ('Positive' is removable by [2.217, 2.343].)

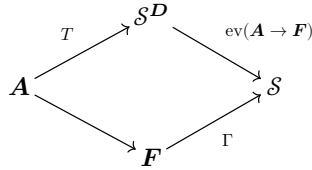
- 1.735 capitalization raises no cardinality unless the category is finite, and positivization preserves infinite cardinalities. any countable (positive) logos has a countable, collectively faithful family of focal representations; a countable logos with coprime terminator is faithfully representable in a countable focal logos.

- 1.74 **GEOMETRIC REPRESENTATION THEOREM** every countable (positive) logos may be faithfully represented in a countable power of the logos of sheaves on the real line; with a coprime terminator, in that logos itself.

- 1.742 for any small (positive) logos $\mathbf{A}$ there is a small category $\mathbf{D}$ and a faithful representation $\mathbf{A} \rightarrow \mathcal{S}^{\mathbf{D}}$. Objects of $\mathbf{D}$: focal representations $\mathbf{A} \rightarrow \mathbf{F}$; morphisms: the focal $\mathbf{F} \rightarrow \mathbf{F}'$ under $\mathbf{A}$.



$$T(\mathbf{A})(\mathbf{A} \rightarrow \mathbf{F}) = \mathbf{F}(1, \mathbf{A}):$$



$\mathbf{D}$ is enlarged until the morphisms out of each of its objects are collectively faithful.

- 1.743 $\mathcal{S}^{\mathbf{D}}$ is focal iff $\mathbf{D}$ has a coterminator; e.g. when $\mathbf{A}$ is focal.

- 1.744 a functor $\mathbf{D}' \rightarrow \mathbf{D}$ induces a representation of pre-logoi $\mathcal{S}^{\mathbf{D}} \rightarrow \mathcal{S}^{\mathbf{D}'}$ by composition, faithful when the functor is onto on objects.

$\mathbf{D}'$ **DOMINATES** $\mathbf{D}$ if some $F : \mathbf{D}' \rightarrow \mathbf{D}$ is onto on objects and **LEFT-FULL**: every $g : F(D) \rightarrow B$ in $\mathbf{D}$ is $F(f)$ for some $f : D \rightarrow A$ in $\mathbf{D}'$. Such F need not be full.

if $\mathbf{D}'$ dominates $\mathbf{D}$ then $\mathcal{S}^{\mathbf{D}}$ is faithfully representable in $\mathcal{S}^{\mathbf{D}'}$ as a logos.

- 1.745 every category (with a coterminator) is dominated by a (rooted) tree: $\mathbf{P}(\mathbf{D})$, the finite nonempty paths of composable morphisms ordered by prolongation on the right, each sent to the target of its last map.

- 1.746 $\mathbf{D}^+$, the tree of finite words of morphisms with the greedy functor $\mathbf{D}^+ \rightarrow \mathbf{P}(\mathbf{D})$, is homogeneous and dominates $\mathbf{D}$. every countable category with a coterminator is dominated by the binary tree 2^* : finite words on two symbols, ordered by prolongation on the right.

- 1.74(10) the **FREYD CURVE** maps $[-1/2, 1/2]$ onto $2^* \cup 2^{\mathbb{N}}$, the finite and infinite binary words with the updeals of the finite words as basic opens.

write $x = \sum_i a_i/3^i$ with $a_i = -1, 0, +1$; delete the zeros to leave $\{a_{n_i}\}_{i=1}^L$, $L \leq \infty$; send x to $\{(-1)^{n_{i-1}} a_{n_i}\}_{i=1}^L$, taking $n_0 = 0$ and reading 2 as $\{-1, +1\}$. Open, continuous and onto, which is what [1.74] needs.

- 1.75 **STONE REPRESENTATION THEOREM** for every small (positive) logos $\mathbf{A}$ there is a space X and a faithful representation $\mathbf{A} \rightarrow \mathcal{S}h(X)$; X may be taken compact, totally disconnected and without isolated points, and for countable $\mathbf{A}$ it may be taken to be the Cantor space, the rationals, or the irrationals.

- 1.751 an object is an **ATOM** if 0 is its unique proper subobject. A logos is **ATOMICALLY BASED** if its atoms form a basis, which forces it boolean; it is **ATOMLESS** if it has no atoms.

the periodic power $\Pi \mathbf{A}$ (the periodic functions $\mathbb{N} \rightarrow \mathbf{A}$) is a sublogos of $\mathbf{A}^{\mathbb{N}}$ with the diagonal $\mathbf{A} \rightarrow \Pi \mathbf{A}$ faithful; it is atomless for non-degenerate $\mathbf{A}$ and stays positive and capital. So: positive capital atomless logoi.

- 1.752 $\mathcal{B}$, the boolean algebra of complemented subterminators of $\mathbf{A}$, is a basis, and atomless when $\mathbf{A}$ is.

the **STONE SPACE** $\hat{\mathcal{B}}$ has the ultrafilters on $\mathcal{B}$ as points and the $\hat{U} = \{\mathcal{F} \mid U \in \mathcal{F}\}$, $U \in \mathcal{B}$, as basic opens; it is compact and totally disconnected.

a boolean algebra is isomorphic to the lattice of **CLOPENS** (sets both open and closed) of its Stone space, so its atoms are the isolated points; $\hat{\mathcal{B}}$ here has none.

- 1.761 $\mathcal{F}$ closed under countable intersection: $\mathbf{A}/\mathcal{F}$ inherits $\mathbf{A}$'s countable unions and coproducts and $\mathbf{A} \rightarrow \mathbf{A}/\mathcal{F}$ preserves them; if closure fails at some size, so does preservation of unions of that size. Closed under arbitrary intersection: $U \in \mathcal{F}$ iff $\bigcap \mathcal{F} \subset U$, and $\mathbf{A}/\mathcal{F} = \mathbf{A}/(\bigcap \mathcal{F})$.

- 1.77 for a binary endo-relation R in a regular category, the **TRANSITIVE CLOSURE** $R^\dagger$ is the minimal transitive relation containing R , and the **TRANSITIVE-REFLEXIVE CLOSURE** R^* the minimal transitive-reflexive one. In a pre-logos each exists iff the other does: $R^* = 1 \cup R^\dagger$ and $R^\dagger = RR^*$.

a **TRANSITIVE LOGOS**, likewise a **TRANSITIVE PRE-LOGOS**, is one in which every endo-relation has a transitive closure.

- 1.771 countable unions in every $\mathcal{S}u\mathcal{B}$, preserved by inverse images (as a logos forces): $R^\dagger = R \cup R^2 \cup \dots \cup R^n \cup \dots$, transitive because composition then distributes with countable union.

- 1.772 a **Σ-TRANSITIVE LOGOS** is a transitive logos in which $R^\dagger$ is the least upper bound of the finite powers of R ; a **Σ-TRANSITIVE PRE-LOGOS** wants that union stable: $R^\dagger = \bigcup_{n=1}^{\infty} R^n$, preserved under inverse images.

σ -transitivity is an infinitary Horn sentence: $(R \subset T) \wedge (R^2 \subset T) \wedge \dots$ implies $R^\dagger \subset T$; for pre-logoi, $f^\#(R^n) \subset A'$ for all n implies $f^\#(R^\dagger) \subset A'$. Its models are closed under products and substructures, and $\mathcal{S}$ is σ -transitive, so representing transitive pre-logoi in a power of $\mathcal{S}$ requires it.

- 1.773 every locally complete regular category is transitive ($R^\dagger$ is the intersection of all transitive relations containing R), but need not be σ -transitive: $R_1 = R$, $R_{\alpha+1} = R_\alpha \cup (R_\alpha)^2$, $R_\beta = \bigcup_{\alpha < \beta} R_\alpha$ at limits.

- 1.775 the **EQUIVALENCE CLOSURE** $R^\equiv$ is the minimal equivalence relation containing R ; in a transitive pre-logos $R^\equiv = (R \cup R^\circ)^*$. An effective regular category has all equivalence closures iff it has coequalizers.

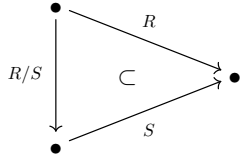
a pre-logos is **E-STANDARD** if every R has an equivalence closure and $R^\equiv$ is always the stable

union of the finite powers of $R^\circ \cup 1 \cup R$. σ -transitive implies E-standard, and E-standardness is necessary for a faithful family of set-valued representations preserving equivalence closures.

- 1.776 every countable positive σ -transitive pre-logos is faithfully representable in a power of $\mathcal{S}$; every countable positive σ -transitive logos in $Sh(X)$ with X the rationals or the irrationals; every countable E-standard bicartesian pre-topos in a power of $\mathcal{S}$.

the last characterizes the countable bicartesian categories representable in a power of $\mathcal{S}$. ‘Positive’ is removable [2.217, 2.343]; ‘countable’ is not.

- 1.78 for relations R, S with a common target, R/S is the maximum relation T , if it exists, with $TS \subset R$: $T \subset R/S$ iff $TS \subset R$. The triangle only semi-commutes, as the containment sign says.



- 1.781 $\text{Rel } x(R/S)y$ iff $\forall_z (ySz) \Rightarrow (xRz)$.
- 1.782 for a map f with the same target as R : R/f exists and equals Rf° .
- 1.783 $(R/S_1)/S_2 = R/(S_2S_1)$: if the left side exists so does the right, and they agree.
- 1.784 in a logos R/S exists for every pair of relations with a common target. Every S is l^r , reducing the question to R/f° ; there $Tf^\circ = (1 \times f)^\#(T)$, so $R/f^\circ = (1 \times f)^{\#\#}(R)$.
- 1.785 conversely, read $A' \rightarrowtail A$ followed by the diagonal $\langle 1, 1 \rangle : A \rightarrow A \times A$ as a relation on A : if A'/f° exists then $f^{\#\#}(A') = 1_B \cap (A'/f^\circ)^\circ(A'/f^\circ)$.
- 1.786 for R, S, T with a common target, $(R/S)(S/T) \subset R/T$; and R/R is transitive and reflexive.
- 1.787 $\bar{R}$ is the minimum reflexive relation with $R\bar{R} \subset \bar{R}$. in a logos $\bar{R} = R^*$: if either exists so does the other, and they are equal.

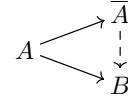
1.8 Adjoint functors, Grothendieck topoi, and exponential categories

- 1.81 **ADJOINT PAIR OF FUNCTORS** $F : \mathbf{A} \rightarrow \mathbf{B}, G : \mathbf{B} \rightarrow \mathbf{A}$: some natural equivalence $(FA, B) \simeq (A, GB)$ exists, natural in both variables — as functors $\mathbf{A}^\circ \times \mathbf{B} \rightarrow \mathcal{S}$. F a **LEFT-ADJOINT** of G , G a **RIGHT-ADJOINT** of F .

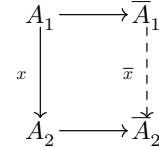
$F \dashv G$	$(FA, B) \simeq (A, GB)$
in posets	$FA \leq B$ iff $A \leq GB$
$\mathbf{A}' \subset \mathbf{A}$ reflective	$(\bar{A}, B) \simeq (A, B), B \in \mathbf{A}'$
$\mathbf{A}' \subset \mathbf{A}$ coreflective	$(B, GA) \simeq (B, A), B \in \mathbf{A}'$
adjoint on the right	$(B, FA) \simeq (A, GB), F, G$ contravariant
adjoint on the left	$(FA, B) \simeq (GB, A), F, G$ contravariant

- 1.811 poset hom-sets have at most one element, so adjointness is $(FA, B) \neq \emptyset$ iff $(A, GB) \neq \emptyset$. Direct image is a left-adjoint of inverse image $f^\#$.

- 1.813 **REFLECTIVE SUBCATEGORY** $\mathbf{A}' \subset \mathbf{A}$: the inclusion has a left-adjoint, whose values are **REFLECTIONS**. Elementary form — every A has $A \rightarrow \bar{A}, \bar{A} \in \mathbf{A}'$, and every $A \rightarrow B, B \in \mathbf{A}'$, factors by a unique $(\bar{A} \rightarrow B) \in \mathbf{A}'$:



one $A \rightarrow \bar{A}$ chosen per A makes reflection a functor: $\bar{x}$ is the unique map with



- 1.814 most reflective subcategories are full; fullness is necessary and sufficient for the reflection functor to be idempotent.
- 1.815 **CLOSURE OPERATION**: the reflection functor of a full reflective subcategory P' of a poset P — order-preserving, idempotent, inflationary. $x \leq y \Rightarrow \bar{x} \leq \bar{y}$, $\bar{\bar{x}} = \bar{x}$, $x \leq \bar{x}$. Conversely its values form a full reflective subcategory.
- 1.816 **COREFLECTIVE** $\mathbf{A}' \subset \mathbf{A}$: the inclusion has a right-adjoint.
- 1.817 G has a left-adjoint F iff every $(A, G(-))$ is representable, by FA ; Fx is then the unique map with

$$\begin{array}{ccc} (A_2, G(-)) & \xrightarrow{\simeq} & (FA_2, -) \\ (x, G(-)) \downarrow & & \downarrow (Fx, -) \\ (A_1, G(-)) & \xrightarrow{\simeq} & (FA_1, -) \end{array}$$

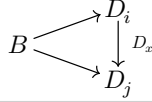
- 1.818 contravariant $F : \mathbf{A} \rightarrow \mathbf{B}, G : \mathbf{B} \rightarrow \mathbf{A}$ are **ADJOINT ON THE RIGHT** if $(B, FA) \simeq (A, GB)$ naturally, **ADJOINT ON THE LEFT** if $(FA, B) \simeq (GB, A)$. Between posets, adjoint on the right = Galois connection.

adjoint on the right iff F then $\mathbf{B} \rightarrow \mathbf{B}^\circ$ is left-adjoint of $\mathbf{B}^\circ \rightarrow \mathbf{B}$ then G ; adjoint on the left iff $\mathbf{A}^\circ \rightarrow \mathbf{A}$ then F is left-adjoint of G then $\mathbf{A} \rightarrow \mathbf{A}^\circ$. Covariant theorems thus yield contravariant ones.

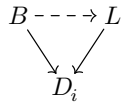
1.82 B^D : objects the functors $D \rightarrow B$, maps the natural transformations; D usually small. **DIAGONAL FUNCTOR** $\Delta : B \rightarrow B^D$: $B \mapsto$ the functor constant at B , $x \mapsto$ the transformation constant at x .

$D \neq \emptyset$: Δ is an embedding, B the subcategory of constant functors. $D = \emptyset$: B^D is the terminator among categories, and Δ has a right-adjoint iff B has a terminator, a left-adjoint iff B has a coterminator.

1.821 objects of B^D are diagrams modelled on D , whose objects i, j, k are the vertices. A map $\Delta(B) \rightarrow D$ is a *lower-bound*: a family $\{B \rightarrow D_i\}_{i \in |D|}$ compatible with every $x : i \rightarrow j$ of D .



D has a coreflection in Δ iff it has a greatest lower-bound $\{L \rightarrow D_i\}$: every lower-bound has a unique map over it, all i .



Discrete D : compatibility evaporates, greatest lower-bound = product diagram.

1.822 **LIMITS** $\lim D$: the values of a right-adjoint of $\Delta : B \rightarrow B^D$. **COLIMITS** $\text{colim } D$: the values of a left-adjoint — least upper bounds.

$\lim \emptyset$ is a terminator, $\text{colim } \emptyset$ a coterminator. For D finitely presented by $\cdot \rightrightarrows \cdot$ (two objects, two non-identity maps) the right-adjoint delivers equalizers, the left-adjoint coequalizers.

1.823 **COMPLETE**: every small diagram has a limit.

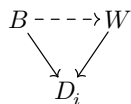
COCOMPLETE: every small diagram has a colimit.

1.824 a family of monics $\{A_i \rightarrowtail A\}_I$ is a diagram modelled on I' (I discrete plus a terminator). Its limit is more than the intersection in $\mathcal{S}u\mathcal{B}(A)$: any $B \rightarrow A$, monic or not, factors through $\lim A_i$ iff it factors through every A_i .

1.825 *complete iff equalizers and arbitrary products dually cocomplete iff coequalizers and arbitrary coproducts and (co-)cartesian iff every finite diagram has a (co-)limit.*

1.827 **CONTINUOUS**: preserves all small limits — from a complete source, iff it preserves equalizers and arbitrary products. **COCONTINUOUS**: preserves all small colimits, from a cocomplete source iff it preserves coequalizers and arbitrary coproducts. A contravariant functor is continuous if it carries colimits to limits.

1.828 *weak-* removes uniqueness. **WEAK-LIMIT** of D : a compatible $\{W \rightarrow D_i\}$ admitting, for every compatible $\{B \rightarrow D_i\}$, at least one

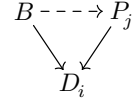


WEAKLY-COMPLETE: every small diagram has a weak-limit.

a functor preserving one weak-limit of a diagram preserves them all; a continuous functor from a complete category preserves all weak-limits.

1.829 avoid “weakly continuous”: a functor that preserves weak-limits preserves limits — preserving weak-limits already forces preservation of monic families.

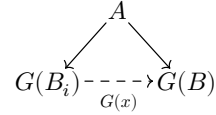
1.82(10) **PRE-LIMIT** for $D : D \rightarrow B$: a set of lower-bounds cofinal in the class of all of them — compatible families $\{\{P_j \rightarrow D_i\}_{i \in |D|}\}_J$ such that every lower-bound $\{B \rightarrow D_i\}$ admits $j \in J$ with



for all $i \in |D|$.

PRE-COMPLETE: every small diagram has a pre-limit. Preserving pre-limits $\Rightarrow$ preserving weak-limits $\Rightarrow$ [1.829] preserving limits.

1.83 **PRE-ADJOINT** for $A \in \mathbf{A}$, given $G : B \rightarrow \mathbf{A}$: a set $\{A \rightarrow G(B_i)\}_I$ such that every $A \rightarrow G(B)$ admits $i \in I$, $x : B_i \rightarrow B$ with



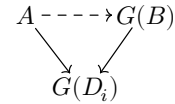
Locally small $\mathbf{A}$: a set $\{B_i\}_I$ in B through one of which every $A \rightarrow G(B)$ factors this way.

B full, G its inclusion: a pre-adjoint is a **PRE-REFLECTION** of A — a set $\{B_i\}$ through one of which every map from A to a B -object factors.

PRE-ADJOINT FUNCTOR: every $A \in \mathbf{A}$ has a pre-adjoint.

THE GENERAL ADJOINT FUNCTOR THEOREM If B is locally small and complete then $G : B \rightarrow \mathbf{A}$ has a left-adjoint iff it is continuous and pre-adjoint.

1.831 **UNIFORMLY CONTINUOUS** $G : B \rightarrow \mathbf{A}$: for every small $D : D \rightarrow B$, G carries the lower-bounds of D to a cofinal family of lower-bounds of $D \xrightarrow{D} B \xrightarrow{G} \mathbf{A}$ — every $\{A \rightarrow G(D_i)\}$ admits a lower-bound $\{B \rightarrow D_i\}$ and

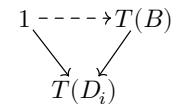


for all i .

such G preserves pre-limits, hence weak-limits, hence limits. B pre-complete: uniformly continuous = preserves pre-limits. B (weakly) complete: = preserves (weak) limits.

THE MORE GENERAL ADJOINT FUNCTOR THEOREM If B is locally small and idempotents split in B then G has a left-adjoint iff it is uniformly continuous and pre-adjoint.

1.832 **POINTWISE CONTINUOUS** $T : B \rightarrow \mathcal{S}$: for every small D , every lower-bound $\{1 \rightarrow T(D_i)\}$ in $\mathcal{S}$ comes from a lower-bound $\{B \rightarrow D_i\}$ in B and a point



G is uniformly continuous iff every $(A, G(-))$ is pointwise continuous.

1.833 smallest subfunctor $T' \subset T$ containing $\{x_i \in T(B_i)\}_I$: $y \in T'(B)$ iff $(Tf)(x_i) = y$ for some $i \in I$, $f : B_i \rightarrow B$. $T' = T$: T is generated by the x_i .

PETTY-FUNCTOR: generated by some set of elements. G is pre-adjoint iff every $(A, G(-))$ is petty.

1.834 **THE GENERAL REPRESENTABILITY THEOREM** If B is locally small and idempotents split in B then $T : B \rightarrow \mathcal{S}$ is representable iff it is a pointwise continuous petty-functor.

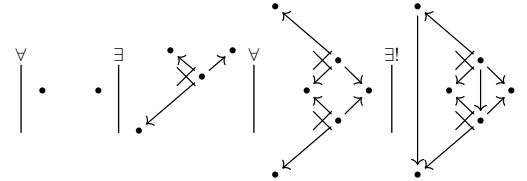
- 1.835 *A locally small category in which idempotents split has a coterminator iff it has a pre-coterminator and all small diagrams have lower-bounds.*
- 1.836 $\hat{B} = \text{Split}(\mathbf{B})$ formally splits every idempotent; $\mathbf{B} \rightarrow \hat{B}$ is full, uniformly continuous and pre-adjoint, with a left-adjoint only where idempotents already split. A set-valued functor from a locally small category is a retract of a representable iff it is a pointwise continuous petty-functor.
- 1.837 $\mathbf{B}$ complete: $\Delta : \mathbf{B} \rightarrow \mathbf{B}^D$ is continuous, and pre-adjoint iff all diagrams modelled on $\mathbf{D}$ have pre-colimits. Hence a complete locally small category is cocomplete iff it is pre-cocomplete.
- 1.838 **WELL-POWERED** $\mathbf{B}$: $\text{Su}\hat{\mathbf{b}}(B)$ is a set for every B . Distinct maps have distinct graphs, so this implies local smallness.
- minimal object*: no proper subobject (an atom is not minimal — it has exactly one). Each object of a complete well-powered category has a unique minimal subobject.
- minimal objects form an initial class $\mathcal{M}$, pre-ordered ($f, g : M \rightarrow A$ have a non-proper equalizer, so $f = g$), every subset of which has a lower bound; an initial set exists iff $\mathcal{M}$ has one. Any map between minimal objects is a cover.
- 1.839 pre-adjoints come from a cardinality function K : objects to cardinals, with $K^{\#(\alpha)}$ a set of isomorphism types for every cardinal α . For continuous $G : \mathbf{B} \rightarrow \mathbf{A}$, $\mathbf{B}$ complete well-powered: B is generated by A through G if $A \rightarrow G(B)$ factors through no $G(B') \subset G(B)$ with $B' \subsetneq B$.
- 1.83(10) **COGENERATING SET** $\{C_i\}_I$: the $\{(-, C_i)\}_I$ are collectively an embedding — $f \neq g : A \rightarrow B$ gives $i \in I$, $h : B \rightarrow C_i$ with $fh \neq gh$. In a complete category: iff every object is embeddable in a product of the C_i .
- THE SPECIAL ADJOINT FUNCTOR THEOREM** If $\mathbf{B}$ is complete, well-powered and has a cogenerating set, and $\mathbf{A}$ is locally small, then every continuous $G : \mathbf{B} \rightarrow \mathbf{A}$ has a left-adjoint.
- A complete well-powered category with a cogenerating set has a coterminator — the minimal subobject of $\Pi_I C_i$.
- 1.83(11) dually: if $\mathbf{A}$ is cocomplete, well-co-powered and has a generating set, every cocontinuous functor from $\mathbf{A}$ to a locally small category has a right-adjoint, and every contravariant continuous functor to a locally small category has an adjoint on the right.
- 1.84 **GIRAUD DEFINITION** of a Grothendieck topos (the first of four): locally small, cocomplete, effective regular, pullbacks preserve arbitrary unions, disjoint unions of objects exist, and there is a generating set. Such is a pre-topos and a logoi.
- 1.842 If $\mathbf{E}$ is a Grothendieck topos then the graphing functor $\mathbf{E} \rightarrow \mathcal{R}\text{el}(\mathbf{E})$ has a right-adjoint. Via 1.843–6 and the special adjoint functor theorem; chapter 2 proves it directly.
- 1.843 A Grothendieck topos is well-powered: a generating set $\{G_i\}_I$ is a basis, every subobject being an equalizer, so $B' \subset B$ is fixed by $\{(G_i, B') \subset (G_i, B)\}_I$. Hence $\mathcal{R}\text{el}(\mathbf{E})$ is locally small.
- comonics coincide with covers, and covers out of A correspond to the equivalence relations on A : a Grothendieck topos is well-copowered.
- 1.844 A Grothendieck topos is locally complete — the union of $\{A_i \rightarrowtail A\}$ is the image of $\Sigma A_i \rightarrow A$, Σ the

disjoint union. Inverse images preserve arbitrary unions, so composition of relations distributes with arbitrary unions.

- 1.845 coproduct $\{u_i : A_i \rightarrow S\}_I$: $u_i u_i^\circ = 1$, $u_i u_j^\circ = 0$ for $i \neq j$, $\bigcup u_i^\circ u_i = 1_S$ — arbitrary coproducts are disjoint unions. A coproduct in $\mathbf{E}$ remains a coproduct in $\mathcal{R}\text{el}(\mathbf{E})$: $R = \bigcup u_i^\circ R_i$ is the unique relation with $u_i R = R_i$, all i .
- 1.846 A coequalizer in $\mathbf{E}$ remains a coequalizer in $\mathcal{R}\text{el}(\mathbf{E})$. A Grothendieck topos is $\mathbf{E}$ -standard: the smallest equivalence relation containing R is $\bigcup_{n=-\infty}^{\infty} R^n$, with $R^0 = 1$, $R^{-n} = (R^\circ)^n$.
- 1.847 Rel the ordering relation on the two-element ordered set is idempotent in $\mathcal{R}\text{el}(\mathcal{S})$ but cannot be split. So $\mathcal{R}\text{el}(\mathbf{E})$ is not cocomplete, though disjoint unions give it coproducts — and, reciprocation making it self-dual, products too.
- 1.85 **EXPONENTIAL** $\mathbf{A}$ with binary products (elsewhere: cartesian closed): every $A \times -$ has a right-adjoint; equivalently every $(A \times -, B)$ is representable, by an object B^A .
- an equivalence $\eta : (-, B^A) \rightarrow (A \times -, B)$ carries $1 \in (B^A, B^A)$ to an **EVALUATION MAP** $e : A \times B^A \rightarrow B$, and every $f : A \times X \rightarrow B$ has a unique $\lambda f : X \rightarrow B^A$ with

$$\begin{array}{ccc} & A \times X & \\ \downarrow 1 \times \lambda f & \searrow f & \\ A \times B^A & \xrightarrow{e} & B \end{array}$$

in the diagrammatic language — the cross marking a product diagram:



- 1.852 a poset is exponential iff it has binary intersections and every a, b have b^a with $x \leq b^a$ iff $a \wedge x \leq b$.
- 1.853 B^A is a bifunctor $\mathbf{A}^\circ \times \mathbf{A} \rightarrow \mathbf{A}$, with $f^A = \lambda \left(A \times B_1^A \xrightarrow{e} B_1 \xrightarrow{f} B_2 \right)$ and $B^g = \lambda \left(A_1 \times B^{A_2} \xrightarrow{g \times 1} A_2 \times B^{A_2} \xrightarrow{e} B \right)$:

$$\begin{array}{ccc} B^{A_2} & \xrightarrow{f^{A_2}} & B_2^{A_2} \\ B_1^g \downarrow & & \downarrow B_2^g \\ B_1^{A_1} & \xrightarrow{f^{A_1}} & B_2^{A_1} \end{array}$$

$(-)^A$ has the left-adjoint $A \times -$, so preserves limits. Commutativity of products makes $B^{(-)}$ adjoint on the right to itself, $(A_1, B^{A_2}) \simeq (A_2 \times A_1, B) \simeq (A_2, B^{A_1})$, hence continuous: colimits to limits. bicartesian exponential: $1^A \simeq 1$, $(B_1 \times B_2)^A \simeq B_1^A \times B_2^A$, $B^0 \simeq 1$, $B^{A_1+A_2} \simeq B^{A_1} \times B^{A_2}$, $(B^{A_1})^{A_2} \simeq B^{A_1 \times A_2}$, $B^1 \simeq B$.

$A \times -$ has a right-adjoint, so preserves colimits: $A \times (B_1 + B_2) \simeq (A \times B_1) + (A \times B_2)$, $A \times 0 \simeq 0$ — the last equivalent to strictness of the coterminator.

caution: 0^A is not 0, and $0^0 \simeq 1$. $(-)^A$ preserves monics, so 0^A is a subterminator: the largest U with $A \times U = 0$.

1.854 in any category with binary products, $\Sigma : \mathbf{A}/B \rightarrow \mathbf{A}$ is a left-adjoint of $\Delta : \mathbf{A} \rightarrow \mathbf{A}/B$: $(\Sigma f, C) \simeq (f, \Delta C)$, with $f : A \rightarrow B$ a slice object and ΔC the projection $B \times C \rightarrow B$.

a right-adjoint of Δ , where it exists, is written $\Pi : \mathbf{A}/B \rightarrow \mathbf{A}$: products of families in $\mathbf{A}$ indexed on B .
If $\Pi : \mathbf{A}/B \rightarrow \mathbf{A}$ exists then A^B exists for every A : $(B \times -, A) \simeq (\Sigma \Delta(-), A) \simeq (\Delta(-), \Delta A) \simeq (-, \Pi \Delta(A))$.

If $\mathbf{A}$ is cartesian and exponential then $\Delta : \mathbf{A} \rightarrow \mathbf{A}/B$ has a right-adjoint Π for every B — for $f : A \rightarrow B$ take the pullback

$$\begin{array}{ccc} \Pi(f) & \xrightarrow{\quad} & A^B \\ \downarrow & \lrcorner & \downarrow f^B \\ 1 & \xrightarrow{\lambda 1_B} & B^B \end{array}$$

$(C, \Pi(f))$, as a subset of $(B \times C, A)$, is then the set of g with

$$\begin{array}{ccc} B \times C & \xrightarrow{g} & A \\ & \searrow f & \swarrow \\ & B & \end{array}$$

which is exactly $(\Delta(C), f)$.

1.856 the slice lemma fails for exponential categories: $\mathbf{P}$ the posets, 3 the three-element chain, $\mathbf{P}/3$ not exponential. $\mathbf{P}/A$ is exponential iff A contains no three-element chain.

1.857 A full coreflective subcategory of an exponential category closed under binary products is exponential: $(A \times C, B) \simeq (C, B^A) \simeq (C, G(B^A))$ for the coreflection G .

EXPONENTIAL IDEAL: a full $\mathbf{A}' \subset \mathbf{A}$ with $B^A \in \mathbf{A}'$ for all $A \in \mathbf{A}$, $B \in \mathbf{A}'$. **REPLETE SUBCATEGORY**: closed under isomorphism type.

A full replete reflective subcategory of an exponential category is an exponential ideal iff reflections preserve products — from $(C, B^A) \simeq (\overline{C}, B^A)$, every $C \rightarrow B^A$ with $B \in \mathbf{A}'$ extends uniquely:

$$\begin{array}{ccc} C & \xrightarrow{\quad} & \overline{C} \\ & \searrow & \swarrow \\ & B^A & \end{array}$$

At $C = B^A$ this forces $C \rightarrow \overline{C}$ to be an isomorphism.

1.858 **KURATOWSKI INTERIOR OPERATION** on a lattice: deflationary, idempotent, intersection-preserving — $x^\bullet \leq x$, $x^\bullet = (x^\bullet)^\bullet$, $(x \wedge y)^\bullet = x^\bullet \wedge y^\bullet$. Its values are the open elements.

LAWVERE-TIERNEY CLOSURE OPERATION (L-T): inflationary, idempotent, intersection-preserving — $x \leq \overline{x}$, $\overline{\overline{x}} = \overline{x}$, $\overline{x \wedge y} = \overline{x} \wedge \overline{y}$. Its values are the closed elements.

1.859 **BASEABLE** B , in a category with binary products: B^A exists — i.e. $(A \times -, B)$ is representable — for every A . The baseable objects form a full subcategory $\mathbf{B}$, with $f^A : B_1^A \rightarrow B_2^A$ defined as if all objects were baseable.

The inclusion $\mathbf{B} \rightarrow \mathbf{A}$ preserves equalizers: the equalizer of f^A, g^A is B_0^A , $\mathbf{B}$ is itself always exponential, since $(B^A)^{A'} \simeq B^{A \times A'}$.

1.9 Topoi

1.9 Any category with pullbacks composes a map with a relation: for $f : A \rightarrow B$ and a table $\langle R; b, c \rangle$ on B, C , pull b back along f .

$$\begin{array}{ccc} R' & \xrightarrow{\quad} & R \xrightarrow{c} C \\ \downarrow & \lrcorner & \downarrow b \\ A & \xrightarrow{f} & B \end{array}$$

$R' \rightarrow A$, $R' \rightarrow C$ is still a monic pair; that table is fR . No regularity needed, and $g(fR) = (gf)R$.

U targeted at C is a **UNIVERSAL RELATION** if every relation R targeted at C is uniquely fU ; that f is written ΛR . A **POWER-OBJECT** of C , written $[C]$, is the source of a universal relation, itself written $\ni \subset [C] \times C$.

$$\begin{array}{ccc} R & \xrightarrow{\quad} & \ni \longrightarrow C \\ \downarrow & \lrcorner & \downarrow \\ A & \xrightarrow{\Lambda R} & [C] \end{array}$$

A **TOPOS** is a cartesian category in which each object has a power-object. A topos is a positive effective transitive logoi, hence a pre-topos.

1.911 $\mathcal{R}el(-, B)$ is contravariant set-valued on any category with pullbacks, $\mathcal{R}el(f, B)$ sending R to fR . $[B]$ exists iff $\mathcal{R}el(-, B) \simeq (-, [B])$. A regular $\mathbf{A}$ is a topos iff the graphing functor $\mathbf{A} \rightarrow \mathcal{R}el(\mathbf{A})$ [1.564] has a right adjoint. In $\mathcal{S}$, $[B]$ is the set of subsets of B .

1.912 $\mathcal{R}el(-, 1) \simeq \mathcal{S}ub(-)$, so $[1]$ — traditionally Ω — is a **SUBJECT CLASSIFIER**. A universal subobject is a monic $\Omega' \rightarrow \Omega$ of which every monic $A' \rightarrow A$ is a unique pullback; χ_A is the **CHARACTERISTIC MAP** of $A' \subset A$, equivalently the unique χ with $\chi^\#(\Omega') = A'$.

$$\begin{array}{ccc} A' & \xrightarrow{\quad} & A \\ \downarrow & \lrcorner & \downarrow \chi_A \\ \Omega' & \xrightarrow{\quad} & \Omega \end{array}$$

Uniqueness of χ_A forces uniqueness of $A' \rightarrow \Omega'$. Taking $A' \rightarrow A$ to be 1_A gives a unique pullback for every A , so each object has exactly one map to Ω' : the universal subobject is a point, henceforth written $1 \xrightarrow{t} \Omega$.

1.913 In any cartesian category with a subobject classifier every subobject is an equalizer — $A' \rightarrow A$ equalizes χ_A and $p_A t$. Consequently covers coincide with epics.

1.914 An n -ary $g : \Omega^n \rightarrow \Omega$ induces $\hat{g}$ on each $\mathcal{S}ub(A)$: $\hat{g}(A_1, \dots, A_n)$ is the subobject with characteristic map $\langle \chi_{A_1}, \dots, \chi_{A_n} \rangle g$. Every operation on $\mathcal{S}ub(-)$ so arises, and each commutes with inverse images: $\hat{g}(f^\# A_1, \dots, f^\# A_n) = f^\# \hat{g}(A_1, \dots, A_n)$.

$g = \chi$ of $\langle t, t \rangle : 1 \rightarrow \Omega \times \Omega$ gives $\hat{g}(A_1, A_2) = A_1 \cap A_2$: Ω is a semi-lattice with unit $1 \xrightarrow{t} \Omega$. $g = \chi$ of $\langle 1, 1 \rangle : \Omega \rightarrow \Omega \times \Omega$ makes $\hat{g}(A_1, A_2)$ the equalizer of χ_{A_1}, χ_{A_2} , so $A' \subset \hat{g}(A_1, A_2)$ iff $A_1 \cap A' = A_2 \cap A'$.

1.919 Every monic endomorphism of Ω is an involution. Absence of non-automorphic monic endomorphisms is a finiteness condition; Ω need not satisfy the dual condition.

1.91(10) A nonempty category with binary products, equalizers and power-objects has a terminator, hence

is a *topos* — the equalizer of $1_{[B]}$ and the constant $\Lambda M_{[B],B}$, where $M_{A,B}$ is the relation tabulated by $A \times B$.

- 1.92 $\Lambda 1 : B \rightarrow [B]$ is the **SINGLETON MAP** in $\mathcal{S}$ it sends b to $\{b\}$. For any $f : A \rightarrow B$, $(f\Lambda 1)\exists = f$ and $f\Lambda 1$ is a map, hence $f\Lambda 1 = \Lambda f$; consequently $\Lambda 1$ is monic.

A topos is exponential. Power-objects are baseable, $(A \times -, [B]) \simeq \mathcal{Su}\ell(- \times A \times B) \simeq (-, [A \times B])$, so $[B]^A$ is constructible as $[A \times B]$; and B is the equalizer of the characteristic map χ of $\Lambda 1$ and of pt , hence baseable [1.859].

$$B \xrightarrow{\Lambda 1} [B] \begin{array}{c} \xrightarrow{\chi} \\ \xrightarrow{pt} \end{array} \Omega$$

- 1.921 $\mathcal{Rel}(-, B) \simeq \mathcal{Su}\ell(- \times B) \simeq (- \times B, \Omega) \simeq (-, \Omega^B)$: $[B]$ is definable as Ω^B . Lawvere's original axioms — bicartesian, exponential, with partial map classifiers [1.934] — follow from power-objects alone.

- 1.922 $[-]$ is the covariant power-object functor, right adjoint of the graphing functor; Ω^- is contravariant. For $g : B_1 \rightarrow B_2$, Ω^g is the unique map inducing $R \mapsto Rg^\circ$ from $\mathcal{Rel}(-, B_2)$ to $\mathcal{Rel}(-, B_1)$; so $\Omega^g = [g^\circ]$.

- 1.923 B^A is cut out of $[A \times B]$ by a pullback.

$$\begin{array}{ccc} B^A & \xrightarrow{\quad} & [A \times B] \\ \downarrow \lrcorner & & \downarrow f \\ 1 & \xrightarrow{\lrcorner A'} & [A] \end{array}$$

In $\mathcal{S}$, f sends $R \subset A \times B$ to $\{a \mid \exists! b, \langle a, b \rangle \in R\}$, and $1 \rightarrow [A]$ names the entire subobject.

- 1.93 *Slice lemma for topoi:* for a topos $\mathbf{E}$ and an object B , $\mathbf{E}/B$ is a topos and $\Delta : \mathbf{E} \rightarrow \mathbf{E}/B$ is a representation of topoi. $\Delta[A]$ is the power-object of $\Delta(A)$; and for $A' \subset A$, $[A']$ is the equalizer of $1_{[A]}$ and the idempotent $e = \Lambda(\exists \cap ([A] \times A'))$.

- 1.931 In any cartesian category $f : A \rightarrow B$ gives $f^\# : A/B \rightarrow A/A$ by pulling back, with left adjoint Σ_f by composing.

$$\begin{array}{ccc} f^\# C & \xrightarrow{\quad} & C \\ \downarrow \lrcorner & & \downarrow f \\ A & \xrightarrow{\quad} & B \end{array}$$

THE FUNDAMENTAL LEMMA OF TOPOI For $f : A \rightarrow B$ in a topos $\mathbf{E}$, $f^\# : \mathbf{E}/B \rightarrow \mathbf{E}/A$ has a right adjoint $\Pi_f : \mathbf{E}/A \rightarrow \mathbf{E}/B$.

- 1.932 $f^\#$ restricted to $\mathcal{Su}\ell(B)$ is inverse image, so Π_f restricted to $\mathcal{Su}\ell(A)$ is the double-sharp operation [1.7]. *The double-sharp axiom holds for topoi.*

- 1.933 *Pullbacks preserve covering families:* for $f : A \rightarrow B$ and a cover $\{C_i \rightarrow B\}_I$, the $\{D_i \rightarrow A\}$ below are a cover.

$$\begin{array}{ccc} D_i & \xrightarrow{\quad} & C_i \\ \downarrow \lrcorner & & \downarrow \\ A & \xrightarrow{\quad} & B \end{array}$$

Consequently a topos is pre-regular.

- 1.934 $\mathcal{P}\mathcal{a}\mathcal{r}(\mathbf{C})$ has the objects of $\mathbf{C}$, and as morphisms $A \rightarrow B$ the pairs $\langle A', f \rangle$ with $A' \subset A$, $f : A' \rightarrow B$; composition of $\langle A', f \rangle$ and $\langle B', g \rangle$ is $\langle f^\#(B'), fg \rangle$.

For a topos $\mathbf{E}$ the inclusion $\mathbf{E} \subset \mathcal{P}\mathcal{a}\mathcal{r}(\mathbf{E})$ has a right adjoint: $\mathcal{P}\mathcal{a}\mathcal{r}(-, A) \simeq (-, \tilde{A})$.

$$\begin{array}{ccc} B' & \xrightarrow{\quad} & B \\ f \downarrow \lrcorner & & \downarrow \\ A & \xrightarrow{\quad} & \tilde{A} \end{array}$$

$\tilde{A} = \Pi_t(A)$; $\tilde{1} = \Omega$, and in a boolean positive logoi $\tilde{A} = A + 1$.

- 1.935 Topoi are pre-regular and satisfy the slice condition [1.541]. *Every topos may be faithfully represented in a capital topos.*

- 1.94 For $F \subset [A]$, $\Gamma(F)$ is a subset of $\mathcal{Su}\ell(A)$: the **SUBOBJECTS NAMED BY F** . The **INTERNALLY DEFINED INTERSECTION** $\cap F$:

$$\begin{array}{ccc} \cap F & \xrightarrow{\quad} & A \\ \downarrow \lrcorner & & \downarrow \\ 1 & \xrightarrow{\quad} & \Omega^F \end{array}$$

$A \rightarrow \Omega^F$ is the adjoint of $F \rightarrow [A] \simeq \Omega^A$, and $1 \rightarrow \Omega^F$ the adjoint of $F \rightarrow 1 \rightarrow \Omega$.

- 1.941 $\cap F$ is preserved by representations of topoi, and is the largest subobject of A that stays a lower bound of the subobjects named by $T(F)$ under every representation T . “Internally defined” is coextensive with being preserved by representations.

- 1.942 The adjoint $1 \rightarrow \Omega^A$ of $\chi_{A'}$ is the **NAME OF A'** , written $\lrcorner A'^\lrcorner$; A' is named by F iff $\lrcorner A'^\lrcorner \subset F$, and the $1 \rightarrow \Omega^F$ of $\cap F$ is the name of the entire subobject. Ω^f is the internally defined inverse image: $\lrcorner B'^\lrcorner \Omega^f = \lrcorner f^\# B'^\lrcorner$.

For $A' \subset A$ named by F , the composite $A \rightarrow \Omega^F$ followed by $\Omega^{\lrcorner A'^\lrcorner}$ is $\chi_{A'}$.

- 1.943 $\cap F$ is contained in every subobject named by F . If F is well-pointed, $\cap F$ is their greatest lower bound, and $B \rightarrow A$ factors through $\cap F$ iff it factors through each subobject named by F .

- 1.944 A topos has a strict coterminal — $\cap \Omega$ in $\mathcal{Su}\ell(1)$, minimal and stably so under every $p^\#$.

- 1.945 A topos is regular. Image of $f : A \rightarrow B$: let $F \subset [B]$ name the B' with $f^\#(B') = A$; then $\cap F = \text{Im } f$.

- 1.946 A topos is a logoi. Binary unions: $F_i \subset [A]$ names the A' containing A_i ; $F = F_1 \cap F_2$ names those containing both, and $\cap F = A_1 \cup A_2$.

- 1.947 A topos is a transitive logoi. For reflexive R on B : F_1 , the equalizer of 1 and $[R \times 1]$ in $[B \times B]$, names the S with $RS \subset S$; F_2 names the reflexive ones; $\cap (F_1 \cap F_2) = R^*$.

- 1.949 The **INTERNALLY DEFINED UNION** $\cup F$ is the direct image $\mu(\exists \cap (F \times A))$ along $\mu : [A] \times A \rightarrow A$. It is an upper bound of every subobject named by F , and the least one when F is well-pointed.

$A' \subset A$ is a permanent lower (upper) bound of F if $T(A')$ stays one under every representation T . $\cap F$ is the greatest permanent lower bound, $\cup F$ the least permanent upper bound.

- 1.94(10) $\Lambda 1 : A \rightarrow [A]$ names the subobjects which are points; $\cup (\Lambda 1)$ is always entire, and is their least upper bound only when A is well-pointed.

A subobject of A is its **WELL-POINTED PART** $A \cdot$ if it is the least upper bound of the points of A , equivalently the greatest well-pointed subobject; no internal construction of $A \cdot$ is possible. A **SOLVABLE TOPOS** is one in which every object has a well-pointed part; there $\mathcal{Su}\ell(A)$ has bounds for every elementarily definable family, but the slice lemma fails. Capital topoi are solvable, so every topos is faithfully represented in a solvable one.

- 1.95 A topos is a pre-topos.

- 1.951 A topos is effective. For an equivalence relation E on A , factor ΛE as $A \xrightarrow{h} B \rightarrow [A]$; then $h^\circ h = 1$ and $hh^\circ = E$.

1.952 A topos is positive.

$$\begin{array}{ccc} 0 & \longrightarrow & 1 \\ \downarrow & \lrcorner & \downarrow \\ A & \xrightarrow{\Lambda 1} & [A] \end{array}$$

since ΛR factors through $\Lambda 1$ iff R is a map, and through $\lrcorner 0$ iff $R = 0$. So $A + 1$ exists; generally $\langle \Lambda 1, \lrcorner 0 \rangle$ and $\langle \lrcorner 0, \Lambda 1 \rangle$ are disjoint monics into $[A] \times [B] \simeq [A + B]$, tabulating $A + B$.

1.954–5 A topos has coequalizers: for $f, g : A \rightarrow B$ put $R = f \circ g$ and $S = (R \cup R^*)^*$; effectiveness makes S the level of some $B \rightarrow C$, the coequalizer. A topos is bicartesian (coequalizers, coterminator, binary coproducts).

1.961 E is an **INJECTIVE** if $(-, E)$ carries monics to epics.

$$\begin{array}{ccc} A & \hookrightarrow & B \\ \downarrow & \dashrightarrow & \downarrow \\ E & & \end{array}$$

In a pre-topos, pushouts of monics are monic [1.651], so E is injective iff every monic $E \hookrightarrow A$ has a right inverse.

E in an exponential category is **INTERNALLY INJECTIVE** if E^- carries monics to epics. In a topos Ω is internally injective: for monic f , $\Omega^f = [f^\circ]$ has left inverse $[f]$, since $\Omega^f[f] = [f^\circ f] = 1$. In a topos Ω is (externally) injective.

1.962 If E is injective in an exponential category then so is E^A , since $(-, E^A) \simeq (- \times A, E)$ and $- \times A$ preserves monics; likewise internally. Hence Ω^A is injective, and $\Lambda 1$ embeds every object in an injective.

1.963 $\tilde{A}$ [1.934] is injective — a partial map $B_1 \rightarrow A$ is a partial map $B_2 \rightarrow A$ whenever $B_1 \hookrightarrow B_2$.

1.964 A category is **VALUE-BASED** if its values form a basis; capital topoi are. In a value-based topos Ω is a cogenerator.

1.965 C in an exponential category **INTERNALLY COGENERATES** if C^- is a contravariant embedding; a cogenerator does. In a topos Ω internally cogenerates.

1.966 A **PROGENITOR** is an object whose subobjects form a generating set. A topos is value-based iff 1 is a progenitor; every Grothendieck topos has one. If G is a progenitor then Ω^G is a cogenerator.

1.967 Equivalent on a locally small topos: (a) arbitrary powers exist; (b) arbitrary copowers exist; (c) arbitrary copowers of 1 exist. Each implies local completeness. (a) says contravariant representables have adjoints, (b) that covariant ones do, (c) that Γ does.

1.968 A locally small topos is complete iff it is cocomplete.

1.969 The **LAWVERE DEFINITION** of a Grothendieck topos: a cocomplete topos with a generating set. The **TIERNEY DEFINITION**: a topos with a progenitor and arbitrary copowers of 1 — equivalently, with a geometric morphism to $\mathcal{S}$.

1.96(11) Slice lemma for Grothendieck topoi: $\mathbf{E}/B$ is a Grothendieck topos, and $\Delta : \mathbf{E} \rightarrow \mathbf{E}/B$ preserves progenitors.

1.971 A is small if it is a subquotient of a **G**-object: $A \leftarrow G' \hookrightarrow G$ with the left arrow a cover. In a boolean Grothendieck topos every object is a coproduct of small objects.

1.972 In a boolean logos with 1 projective, 1 is a progenitor. Equivalent on a Grothendieck topos: (1)

boolean and 1 projective; (2) boolean and value-based; (3) boolean with a choice progenitor; (4) every object a coproduct of sub-terminators; (5) AC.

1.973 A topos is **IAC** (internal axiom of choice) if $(-)^A$ [1.853] preserves epics for every A .

1.974 A topos is AC iff it is IAC and 1 is projective. Every functor from an AC topos preserves epics, since it preserves left-invertibility.

1.976 A small topos may be faithfully represented in an AC topos iff it is IAC — take $C_A \subset A^{[A]^+}$ naming the choice maps contained in the universal relation.

1.977 Any universal sentence in the predicates of topoi which follows from AC follows from IAC. In particular an IAC topos is boolean.

1.979 For any topos $\mathbf{A}$ there is a boolean topos $\mathbf{B}$ and a faithful bicartesian representation $\mathbf{A} \rightarrow \mathbf{B}$ [2.542].

1.98 A **NATURAL NUMBERS OBJECT** (NNO) is $1 \xrightarrow{0} N \xrightarrow{s} N$ such that every $1 \xrightarrow{a} A \xrightarrow{t} A$ admits a unique $N \rightarrow A$:

$$\begin{array}{ccccc} 1 & \xrightarrow{0} & N & \xrightarrow{s} & N \\ & \searrow a & \downarrow & \downarrow & \downarrow \\ & & A & \xrightarrow{t} & A \end{array}$$

Unique up to isomorphism. In $\mathcal{S}$: zero and successor. The topos of finite sets has no NNO.

1.981 For a NNO and any $A \xrightarrow{x} B \xrightarrow{t} B$ there is a unique $A \times N \rightarrow B$:

$$\begin{array}{ccccc} A & \xrightarrow{\langle 1_A, 0 \rangle} & A \times N & \xrightarrow{1_A \times s} & A \times N \\ & \searrow x & \downarrow & \downarrow & \downarrow \\ & & B & \xrightarrow{t} & B \end{array}$$

— transpose x, t into $1 \rightarrow B^A \rightarrow B^A$.

1.983 Recursion: for a NNO and any $g : A \rightarrow B$, $h : A \times N \times B \rightarrow B$ there is a unique $f : A \times N \rightarrow B$ with $\langle 1_A, 0 \rangle f = g$ and $(1_A \times s)f = \langle p_1, p_2, f \rangle h$.

1.984 Addition, multiplication and exponentiation on a NNO by the usual recursion equations: $\langle 1_N, 0 \rangle a = 1_N$, $(1_N \times s)a = as$; $\langle 1_N, 0 \rangle m = 0$, $(1_N \times s)m = \langle p_1, m \rangle a$; $\langle 1_N, 0 \rangle e = 0s$, $(1_N \times s)e = \langle p_1, e \rangle m$.

1.985 For a NNO, $1 \xrightarrow{0} N \xleftarrow{s} N$ is a coproduct and $N \rightrightarrows N \rightarrow 1$ (by 1_N and s) is a coequalizer.

$$\begin{array}{ccc} & 1 & N \\ & \searrow & \swarrow \\ & 0 & N \\ & \swarrow & \searrow \\ N & \xrightarrow{1_N} & N \end{array}$$

1.986 Given $1 \xrightarrow{a} A$ and $A \xrightarrow{f} A$ there is $A' \subset A$ carrying a', f' with

$$\begin{array}{ccccc} 1 & \xrightarrow{a'} & A' & \xrightarrow{f'} & A' \\ & \searrow a & \downarrow & \downarrow & \downarrow \\ & & A & \xrightarrow{f} & A \end{array}$$

and with $\binom{a'}{f'} : 1 + A' \rightarrow A$ a cover — in $\mathcal{R}\mathcal{E}(\mathbf{A})$ take $A'' = af^*$, $f'' = A''f$.

1.987 $1 \xrightarrow{a} A, A \xrightarrow{t} A$ has the **PEANO PROPERTY** iff every subobject $B \subset A$ that allows a and allows $t \upharpoonright B$, the composite $B \subset A \xrightarrow{t} A$, is entire. The A' of 1.986 is the least subobject allowing both a and $A' \subset A \xrightarrow{f} A$, and it has the Peano property.

1.988 In a boolean topos, if $1 \xrightarrow{a} A \xleftarrow{t} A$ is a coproduct and $A \rightrightarrows A \rightarrow 1$ (by 1_A and t) a coequalizer, then $1 \xrightarrow{a} A \xrightarrow{t} A$ has the Peano property. (“Boolean” is removable by [2.542].)

1.989 Under the hypotheses of 1.988, if $(\begin{smallmatrix} c \\ u \end{smallmatrix}) : 1 + C \rightarrow C$ is a cover and $cf = a$, $uf = ft$, then $f : C \rightarrow A$ is monic.

1.98(10) Converse of 1.985: in any topos, if $1 \xrightarrow{a} A \xleftarrow{t} A$ is a coproduct and $A \rightrightarrows A \rightarrow 1$ a coequalizer, then $1 \xrightarrow{a} A \xrightarrow{t} A$ is a NNO. Uniqueness from the Peano property applied to the equalizer of two candidates; existence from 1.986 applied to $\langle a, b \rangle$ and $t \times v$.

1.98(11) A bicartesian functor $T : \mathbf{A} \rightarrow \mathbf{A}'$ of topoi carries a NNO to a NNO — by the bicartesian characterization [1.985, 1.98(10)].

1.98(12) An **A-ACTION** is an object B with $1 \xrightarrow{b} B$ and $A \times B \xrightarrow{v} B$. A **FREE A-ACTION** is an A -action $A \times 1 \xrightarrow{1_A \times e} A \times A^* \xrightarrow{s} A^*$ universal among them:

$$\begin{array}{ccccc} A \times 1 & \xrightarrow{1_A \times e} & A \times A^* & \xrightarrow{s} & A^* \\ & \searrow & \downarrow 1_A \times f & \downarrow f & \downarrow \\ & & A \times B & \xrightarrow{v} & B \end{array}$$

Unique up to isomorphism; a NNO is a free 1-action. In $\mathcal{S}$, A^* is the set of finite lists, e the empty list, s prefixing.

1.98(13) $A \times 1 \rightarrow A \times A^* \rightarrow A^*$ is a free A -action iff $(\begin{smallmatrix} e \\ s \end{smallmatrix}) : 1 + A \times A^* \rightarrow A^*$ is iso and $A \times A^* \rightrightarrows A^* \rightarrow 1$ is a coequalizer.

1.98(14) In a topos with a NNO every object A has a free A -action — take A^* the least subobject of $(1 + A)^N$ allowing e and $A \times A^* \subset A \times (1 + A)^N \rightarrow (1 + A)^N$.